

THE SUBREGULAR AND SUBMAXIMAL P-CELLS

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ABSTRACT. Cells for the canonical and p -canonical bases organise the representation theory and geometry of Hecke categories. We determine the relevant p -canonical basis elements and the resulting p -cell structure for the subregular and submaximal cells in all classical types.

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1. INTRODUCTION

Kazhdan–Lusztig theory is one of the main bridges between the geometry of Schubert varieties, combinatorics and representation theory. Let G be a connected reductive complex algebraic group, let $B \subset G$ be a Borel subgroup, and let W be its Weyl group. The flag variety G/B is stratified by Schubert cells

$$X_w^\circ = BwB/B, \quad w \in W,$$

and their closures X_w are the Schubert varieties. Over a field of characteristic zero, the intersection cohomology complexes of the Schubert varieties have characters given by the Kazhdan–Lusztig (KL) basis b_w of the Hecke algebra; see [KL1, KL2].

The situation changes over a field of characteristic $p > 0$. The decomposition theorem for maps such as Bott–Samelson resolutions can fail with modular coefficients: torsion in the corresponding integral intersection cohomology, or equivalently a reduction in rank of the integral intersection forms after reduction modulo p , can change the indecomposable summands. Parity sheaves provide a robust replacement: they are designed to retain the useful decomposition-theoretic features of intersection cohomology in this modular setting. As it turns out, for each $w \in W$, there is an indecomposable parity sheaf supported on X_w , and its character is the p KL basis element pb_w ; see [JMW] and [JW]. Thus an inequality

$${}^pb_w \neq b_w$$

records a genuine difference between the geometry of X_w in characteristic p and its characteristic zero geometry.

One way to see this difference is through a Bott–Samelson resolution of X_w . The decomposition of the corresponding Bott–Samelson object is controlled by integral local intersection forms. If the rank of one of these forms decreases when reducing modulo p , an additional indecomposable summand appears, and hence the p KL basis differs from the KL basis. In this sense, the coefficients of the p KL basis detect modular singularities of Schubert varieties.

Although fairly new, the study of the p KL basis and its associated structures has developed rapidly. Williamson’s torsion explosion showed that such modular phenomena occur for primes far larger than previously expected and ruled out several optimistic forms of long-standing conjectures in modular representation theory; see [Wi]. At the same time, the p KL basis has become an effective tool for modular character theory. It occurs in character formulas for tilting modules and, through modular Koszul duality and Smith theory, in character formulae for representations of reductive algebraic groups; see, for example, [RW1, AMRW, RW2].

There has also been substantial progress in computation and structure. Algorithms now make it possible to calculate large portions of the p KL basis without performing the entire calculation directly in a combinatorial object called the diagrammatic Hecke category; see [GJW]. Nevertheless, explicit uniform formulas

remain rare and are largely confined to particularly structured situations; see, for example, [Ba, BDHN]. At the opposite extreme, recent asymptotic results show that disagreement between the KL and p KL bases is widespread in the classical families; see [BT].

Thus there is a substantial gap between knowing that modular phenomena are widespread and being able to describe their occurrences explicitly.

1.1. Cells: Why care? The multiplication of canonical basis elements defines the ordinary left, right and two-sided KL cells. Replacing the canonical basis by the p KL basis defines the corresponding p -cells. Geometrically, these relations measure which indecomposable parity sheaves can occur from others under convolution. Categorically, they describe the cell structure of the geometric or diagrammatic Hecke category. Thus p -cells record how the modular changes in the p -canonical basis reorganize the multiplication structure of the Hecke category.

Cells connect Hecke theory to several neighboring parts of representation theory and geometry. At the decategorified level, they organize cell modules, Lusztig families, special representations and primitive ideals; see, e.g., [KL1, Lu1, Jo]. Geometrically, two-sided cells also give rise to asymptotic Hecke algebras and monoidal categories defined using truncated convolution; see, e.g., [Lu2, Be, ERT]. Finally, cells control tensor ideals for quantum tilting modules, while antispherical p -cells are expected to play an analogous role for tilting modules in positive characteristic and their support varieties; see, e.g., [Os, AHR1, AHR2].

Cell structure is also fundamental in 2-representation theory. The indecomposable objects of the Hecke category are the indecomposable 1-morphisms of a finitary 2-category, and its left, right and two-sided cells organize their behavior under composition; see [MM1]. Every simple 2-representation has a unique maximal two-sided cell which acts nontrivially, called its apex. The problem of classifying simple 2-representations therefore breaks naturally into separate problems, one for each possible apex; see, for example, [MM2, MMTZ1]. Over a field of characteristic zero, this approach led to a proof that the Hecke categories of finite Coxeter type admit only finitely many equivalence classes of simple 2-representations, and to their classification in every finite type except H_3 and H_4 ; see [MMMTZ2].

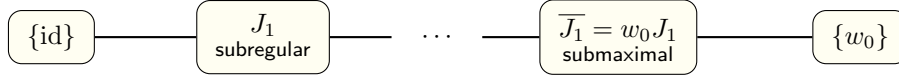
In positive characteristic, the indecomposable objects, their multiplication and hence their cells may change. Before one can attempt an analogous result, one must first understand the new two-sided p -cells which can occur.

In finite type A , the p -cells are completely understood: they agree with the ordinary KL cells for every prime and hence admit the usual description via the Robinson–Schensted correspondence; see [Je1]. The same work shows that in finite types B and C , for $p > 2$, ordinary KL cells decompose into p -cells. Moreover, the p -canonical basis in type A is cellular; see [Je2]. Further structural

and low-rank results appear in [EJ]. The behavior of p -cells under parabolic induction has also been studied; see [JP].

Nevertheless, there are few uniform descriptions of p -cell structures outside type A . The aim of the present paper is to give such descriptions for the first nontrivial cells at the two ends of the ordinary two-sided cell order.

1.2. Contribution of this article. We study the first nontrivial cells at the two ends of the ordinary two-sided cell order:



Here J_1 is the ordinary subregular cell, consisting of the non-identity elements with a unique reduced expression, and w_0 denotes the longest element. The submaximal cell $\overline{J}_1 = w_0 J_1$ is its translate to the opposite end of the cell order.

These cells are natural first test cases. The elements in the ordinary subregular cell are well-understood and have an explicit description in terms of paths in the Coxeter graph. At the same time, they are large enough to display genuinely modular phenomena.

Our main result is the following.

Main result. In Theorems 3.11 and 4.10, we determine the two-sided p -cell decompositions of the subregular and submaximal cells in all classical types A, B, C, D . Except in types B_n and C_n in characteristic two, the ordinary cells remain single two-sided p -cells. In characteristic two they split, and the answers for types B_n and C_n are different.

We also determine the subregular p -cells in all exceptional types and the submaximal p -cells in types G_2 and F_4 (Theorem 5.2). The exceptional primes are 2 and 3 in type G_2 , and 2 in type F_4 . In types E_6, E_7, E_8 , the subregular cell remains a single two-sided p -cell for every prime.

In each case, we further record which p -cells contain the longest element of a parabolic subgroup.

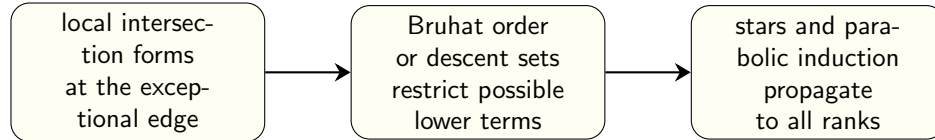
As far as we know, these are the first uniform descriptions in arbitrary rank of nontrivial two-sided p -cells outside type A . Moreover, we give more than the decomposition into two-sided p -cells. We determine the relevant p KL basis elements in classical types explicitly in Theorems 3.7 and 4.4, with one isolated submaximal element in type C_n where a complete expansion is not needed for the cell calculation. The exceptional cases are treated computationally in Section 5.

1.3. The proof in a nutshell. Let us briefly explain the proof. The first step is local and geometric. We compute the intersection forms controlling the decomposition of selected Bott–Samelson objects in small rank. The only reductions

in rank in the classical families occur at the exceptional edge and in characteristic two. These calculations give the small rank results from which all remaining formulae are obtained, with the exception of one case in type C_n .

The second step is combinatorial. Which canonical basis element can occur as a lower term of a p KL basis element is severely constrained by the Bruhat order in the subregular case and the left and right descent sets in the submaximal case. These restrictions leave very few possibilities, which turns what could be a large calculation in the Hecke category into a small list of local cases.

The third step propagates the local information through the Coxeter graph. Inversion exchanges left and right calculations. Star operations move the rank two relations along the simply laced part of the graph, while induction from parabolic subgroups transports calculations from lower rank to arbitrary rank. The proof can be summarized as follows:



Once the necessary p KL expansions are known, the p -cell relations are obtained from those in small rank by using star operations and parabolic induction, and counting the intersections of left and right cells determines the final cell shapes. Thus the calculation is mostly uniform in rank and ultimately rests on a small amount of local geometry. Exceptional types are treated computationally.

1.4. Structure of this article. The paper is organized as follows. In Section 2 we recall the p KL basis and collect the tools used throughout the paper. Section 3 treats the subregular cell in classical type, first determining the relevant p KL basis elements and then the resulting p -cells. Section 4 follows the same program for the submaximal cell. Finally, Section 5 contains the calculations in exceptional type.

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2. BACKGROUND

2.1. The p -canonical basis and p -cells.

2.1.1. Coxeter systems. Let (W, S) be a *Coxeter system*. In other words, there is a matrix $(m_{s,t})_{s,t \in S}$ with $m_{s,s} = 1$ and $m_{s,t} = m_{t,s} \in \{2, 3, \dots\} \cup \{\infty\}$ for $s \neq t$ such that

$$W = \langle s \in S \mid (st)^{m_{s,t}} = 1 \rangle.$$

Here we will assume that W is finite and $m_{s,t} \in \{1, 2, 3, 4, 6\}$ for $s, t \in S$, that is, (W, S) is a crystallographic Coxeter system.

We also fix a Cartan realisation $\mathfrak{h}$ of (W, S) over $\mathbb{Z}$ (see [AMRW, Section 10.1]). In particular, it is a free $\mathbb{Z}$ -module of finite rank with an action of W given on the *simple co-roots* $\{\alpha_s^\vee\}_{s \in S}$ by

$$t.\alpha_s^\vee = \alpha_s^\vee - \alpha_t(\alpha_s^\vee)\alpha_t^\vee$$

for $t \in S$. Here $\alpha_t \in \text{Hom}(\mathfrak{h}, \mathbb{Z})$ is a *simple root*.

To (W, S) one associates a *Dynkin diagram* as follows. Its vertices are labelled by S . For $s, t \in S$ with $s \neq t$, the corresponding vertices are joined by $|\alpha_s(\alpha_t^\vee)|$ edges if $|\alpha_s(\alpha_t^\vee)| \geq |\alpha_t(\alpha_s^\vee)|$. If moreover, $|\alpha_s(\alpha_t^\vee)| > 1$, the edge is oriented towards the vertex labelled s .

We use the following labelling conventions for Dynkin diagrams in classical types:

$$\begin{array}{ll} A_n : \textcircled{1} - \textcircled{2} - \textcircled{3} - \cdots - \textcircled{n-1} - \textcircled{n} & B_n : \textcircled{1} \leftarrow \textcircled{2} - \textcircled{3} - \cdots - \textcircled{n-1} - \textcircled{n} \\ C_n : \textcircled{1} - \textcircled{2} \Rightarrow \textcircled{3} - \textcircled{4} - \cdots - \textcircled{n-1} - \textcircled{n} & D_n : \begin{array}{c} \textcircled{1} \\ | \\ \textcircled{2} - \textcircled{3} - \textcircled{4} - \cdots - \textcircled{n-1} - \textcircled{n} \end{array} \end{array}$$

The displayed diagrams are schematic: in type A_n, B_n, C_n the vertices are $1, \dots, n$ from left to right. In types B_n and C_n , the exceptional edge is between s_1 and s_2 . In type D_n , the fork is attached at s_3 .

For exceptional types, we use the following labelling convention:

$$\begin{array}{lll} G_2 : \textcircled{1} \leftarrow \textcircled{2} & F_4 : \textcircled{1} - \textcircled{2} \Rightarrow \textcircled{3} - \textcircled{4} \\ E_6 : \begin{array}{c} \textcircled{4} \\ | \\ \textcircled{1} - \textcircled{3} - \textcircled{5} - \textcircled{6} \\ | \\ \textcircled{2} \end{array} & E_7 : \begin{array}{c} \textcircled{4} \\ | \\ \textcircled{1} - \textcircled{3} - \textcircled{5} - \textcircled{6} - \textcircled{7} \\ | \\ \textcircled{2} \end{array} & E_8 : \begin{array}{c} \textcircled{4} \\ | \\ \textcircled{1} - \textcircled{3} - \textcircled{5} - \textcircled{6} - \textcircled{7} - \textcircled{8} \\ | \\ \textcircled{2} \end{array} \end{array}$$

2.1.2. Expressions, decoration, defect. The elements in S are called *simple reflections* and an *expression* $(s_1, \dots, s_n)$ is a sequence of elements in S . The product $s_1 \cdots s_n$ gives an element $w \in W$ and we will write $\underline{w} = (s_1, \dots, s_n)$ (or sometimes $\underline{w} = s_1 \cdots s_n$). We say that the expression is *reduced* if it has minimal length amongst all expressions for w . The length of a reduced expression defines the *length* $l(w)$ of the element w .

Convention 2.1 (Shortest path). Let W be of classical type, with simple reflections $S = \{s_1, \dots, s_n\}$. If a, b are vertices of the Coxeter graph and there is a unique shortest path from a to b , write

$$[a, b]$$

for the product of the simple reflections along this path, read from a to b . Thus, in type A_n ,

$$[a, b] = \begin{cases} s_a s_{a+1} \cdots s_b, & a \leq b, \\ s_a s_{a-1} \cdots s_b, & a \geq b. \end{cases}$$

The same notation is used in type D_n , on the forked Coxeter graph with the fork attached at s_3 .

Convention 2.2 (Exceptional truncations). Assume W is of type B_n or C_n , with exceptional edge between s_1 and s_2 . For $a, b \geq 2$, write

$$[a, 1, b] = s_a s_{a-1} \cdots s_2 s_1 s_2 \cdots s_b.$$

For $\min(a, b) \geq k \geq 1$, define the k -th truncation of this element by

$$[a, k, b] := s_a s_{a-1} \cdots s_{k+1} s_k s_{k+1} \cdots s_b.$$

Thus $[a, 1, b]$ is the original word, while larger k means that the word turns around earlier, before reaching the exceptional edge.

For $\underline{w} = (s_1, \dots, s_n)$ an expression, a *subexpression* of $\underline{w}$ is a sequence $e = (e_i)_{1 \leq i \leq n} \in \{0, 1\}^n$. If $\underline{w}$ is a reduced expression for w , then we write that $x = s_1^{e_1} \cdots s_n^{e_n} \leq w$. This defines the *Bruhat order* on W , see, e.g., [EMTW, Section 1.2.6] for more details.

To a subexpression e of $\underline{w}$, one associates a *decoration* $d = (d_i)_{1 \leq i \leq n} \in \{U, D\}^n$ by setting

$$d_1 = U \text{ and } d_i = \begin{cases} U & \text{if } s_1^{e_1} \cdots s_{i-1}^{e_{i-1}} s_i > s_1^{e_1} \cdots s_{i-1}^{e_{i-1}} \\ D & \text{otherwise} \end{cases} \text{ for } 2 \leq i \leq n.$$

The defect of e is given by

$$\text{df}(e) := |\{1 \leq i \leq n \mid d_i e_i = U0\}| - |\{1 \leq i \leq n \mid d_i e_i = D0\}|.$$

Example 2.3. Assume that W is of type B_6 .

Let $\underline{w} = s_6 s_5 s_4 s_3 s_2 s_1 s_2 s_3 s_4 s_5 s_6 = [6, 1, 6]$ and $e = 111010101$. It corresponds to the element $s_6 s_5 s_4 s_2 s_2 s_4 s_6 = s_5 s_6 s_5$. Its decoration is

$$U1U1U1U0U1U0D1U0D1D0U1$$

and $\text{df}(e) = 2$.

2.1.3. Hecke algebra. To (W, S) we associate the *Hecke algebra* H , which is free over $\mathbb{Z}[v, v^{-1}]$ and is generated by $\{\delta_w \mid w \in W\}$, subject to relations for $s, t \in S$

$$\begin{aligned} \delta_s \delta_s &= (v^{-1} - v) \delta_s + 1, \\ \underbrace{\delta_s \delta_t \delta_s \cdots}_{m_{s,t}} &= \underbrace{\delta_t \delta_s \delta_t \cdots}_{m_{s,t}}. \end{aligned}$$

For $w \in W$ with reduced expression $(s_1, \dots, s_n)$, we set $\delta_w := \delta_{s_1} \cdots \delta_{s_n}$. The set $\{\delta_w \mid w \in W\}$ is the standard basis.

There is a unique $\mathbb{Z}$ -linear involution $(\bar{})$ on H satisfying $\bar{v} = v^{-1}$ and $\bar{\delta_x} = \delta_{x^{-1}}$.

We fix

$$b_s = \delta_s + v.$$

Observe that $b_s = \bar{b_s}$. For an expression $\underline{w} = (s_1, \dots, s_n)$, we write $b_{\underline{w}} = b_{s_1} \cdots b_{s_n}$.

2.1.4. *Hecke category and p -canonical basis.* Let O be any complete local ring. Consider $\mathfrak{h}_O = \mathfrak{h} \otimes_{\mathbb{Z}} O$ and assume it satisfies Demazure surjectivity. The main reference is [EW], more precisely Lemma 6.24, Theorem 6.25, and Corollary 6.26.

Let $\mathcal{H}$ be the diagrammatic Hecke category associated with $\mathfrak{h}_O$.

Theorem 2.4 (Properties of $\mathcal{H}$). *Let O be a complete, local, integral domain. Then*

- (1) $\mathcal{H}$ is a Krull-Schmidt category.
- (2) For all $w \in W$ there exists a unique indecomposable object ${}^O B_w \in \mathcal{H}$ which is a direct summand of $B_{\underline{w}}$ for any reduced expression $\underline{w}$ of w and which is not isomorphic to a grading shift of any direct summand of any expression $B_{\underline{v}}$ for $v < w$. In particular, the object ${}^O B_w$ does not depend up to isomorphism on the reduced expression $\underline{w}$ of w .
- (3) The set $\{{}^O B_w \mid w \in W\}$ gives a complete set of representatives of the isomorphism classes of indecomposable objects in $\mathcal{H}$ up to grading shift.
- (4) There exists a unique isomorphism of $\mathbb{Z}[v, v^{-1}]$ -algebras $\text{ch} : [\mathcal{H}] \rightarrow H$ sending $[{}^O B_s]$ to b_s for all $s \in S$, where $[\mathcal{H}]$ denotes the split Grothendieck group of $\mathcal{H}$.

Note that in particular, we have $\text{ch}(B_{\underline{w}}^O) = b_{\underline{w}}$ for any $w \in W$. We are now ready to define the p -canonical basis as in [JW, Definition 3.2].

Definition 2.5 (p -canonical basis). Let $\mathbb{k}$ be a field of positive characteristic $p > 0$. The p -canonical basis of H is $\{{}^p b_w \mid w \in W\}$ where

$${}^p b_w := \text{ch}([{}^{\mathbb{k}} B_w]).$$

If $\mathbb{K}$ is a field of characteristic zero, the map ch yields the canonical basis $\{b_w \mid w \in W\}$.

Remark 2.6. The p -canonical basis depends only on the characteristic of $\mathbb{k}$ and of the choice of the realisation $\mathfrak{h}$.

The map ch yields the various coefficients. We fix the following notation for $x \in W$ with reduced expression $\underline{x}$. We write

$$b_{\underline{x}} = b_x + \sum_{z < x} h_{z,x} b_z, \quad \text{and} \quad b_{\underline{x}} = {}^p b_x + \sum_{z < x} {}^p h_{z,x} {}^p b_z$$

and

$${}^p b_x = b_x + \sum_{y < x} {}^p m_{y,x} b_y.$$

In particular, we have

$$(2.1) \quad {}^p b_x = b_x + \sum_{z < x} b_z (h_{z,x} - {}^p h_{z,x} - \sum_{z < y < x} {}^p m_{z,y} {}^p h_{y,x}).$$

By [JW, Theorem 4.2], $h_{z,x}, {}^p h_{z,x}, {}^p m_{z,x} \in \mathbb{Z}_{\geq 0}[v, v^{-1}]$.

Let $x, y \in W$. We write

$$b_x b_y = \sum_{z \in W} \mu_{x,y}^z b_z \quad \text{and} \quad {}^p b_x {}^p b_y = \sum_{z \in W} {}^p \mu_{x,y}^z {}^p b_z.$$

For $z \in W$, the coefficient of b_z in ${}^p b_x {}^p b_y$ is then given by

$$(2.2) \quad \sum_{v,w} {}^p m_{v,x} {}^p m_{w,y} \mu_{v,w}^z = {}^p \mu_{x,y}^z + \sum_{w > z} {}^p m_{z,w} {}^p \mu_{x,y}^w.$$

We say that $x \leq_L^p y$ (resp. $x \leq_R^p y$) if there is $w \in W$ such that ${}^p \mu_{w,x}^y \neq 0$ (resp. ${}^p \mu_{x,w}^y \neq 0$). Note that this is the opposite preorder to that given in [Je1, Definition 3.1]. The partial preorder $\leq_J^p$ is generated by $\leq_L^p$ and $\leq_R^p$, i.e. $x \leq_J^p y$ if there is $w, w' \in W$ such that ${}^p b_y$ appears as a summand of ${}^p b_w {}^p b_x {}^p b_{w'}$. The equivalence classes for each of the partial preorders are called left, right or two-sided p -cells.

If $p = 0$, we will drop the superscript 0 and write $\leq_J$ instead of $\leq_J^0$ for instance.

2.2. Useful tools to compute the p -canonical basis. We collect various useful results to compute the p -canonical basis that we will apply later on.

2.2.1. Local intersection form. There is a general method to compute the coefficients ${}^p h_{y,x}$ using only the information at the level of $B_{\underline{x}}$. We give a quick overview, referring the reader to [GJW, Section 5] or [EMTW, Section 27.3] for more details.

Let $\underline{x}, \underline{y}$ be two reduced expressions in W such that $y \leq x$ (in the Bruhat order). Let $\text{Hom}_{<\underline{y}}^\bullet(B_{\underline{y}}, B_{\underline{x}})$ the set of morphisms in the graded space $\text{Hom}^\bullet(B_{\underline{y}}, B_{\underline{x}})$ that factors through some $B_{\underline{w}}$ for $\underline{w}$ a reduced expression of $w < y$. Set $\text{Hom}_{\not<\underline{y}}^\bullet(B_{\underline{y}}, B_{\underline{x}}) := \text{Hom}^\bullet(B_{\underline{y}}, B_{\underline{x}}) / \text{Hom}_{<\underline{y}}^\bullet(B_{\underline{y}}, B_{\underline{x}})$.

For any $d \in \mathbb{Z}$, the d -th graded piece of the local intersection form is the restriction

$$I_{\underline{x}, \underline{y}}^d : \text{Hom}_{\not<\underline{y}}^d(B_{\underline{y}}, B_{\underline{x}}) \times \text{Hom}_{\not<\underline{y}}^{-d}(B_{\underline{x}}, B_{\underline{y}}) \rightarrow \mathbb{Z}.$$

We then have

$${}^p h_{y,x} = \sum_{d \in \mathbb{Z}} \text{rank}(I_{\underline{x}, \underline{y}}^d \otimes_O \mathbb{k}) v^d.$$

Remark 2.7. The spaces $\text{Hom}_{\not<\underline{y}}^\bullet(B_{\underline{y}}, B_{\underline{x}})$ and $\text{Hom}_{\not<\underline{y}}^\bullet(B_{\underline{x}}, B_{\underline{y}})$ have basis given in terms of *light leaves*. Let e be a subexpression of $\underline{x}$ such that $\underline{x}^e = y$. A light leaf is a particular morphism from $B_{\underline{x}}$ to $B_{\underline{x}^e}$, which can be given in terms of diagram, using the decoration of e . Its degree is given by the defect of e . For more information on how to compute light leaves, see [EMTW, Section 10.4].

2.2.2. *Descent sets.* One first tool to control the decomposition of the p -canonical basis element ${}^p b_x$ into b_y is to use the descent sets of x .

Convention 2.8 (Descent sets). Let $w \in W$. The *left descent set* $L(w)$ of w is defined as follows:

$$L(w) := \{s \in S \mid l(sw) < l(w)\}.$$

The *right descent set* $R(w)$ is defined symmetrically as

$$R(w) := \{s \in S \mid l(ws) < l(w)\}.$$

Note that $L(w) = R(w^{-1})$. Moreover, for $x, y \in W$,

$$\begin{aligned} x \leq_L^p y &\implies R(x) \subseteq R(y) \\ x \leq_R^p y &\implies L(x) \subseteq L(y). \end{aligned}$$

Lemma 2.9 ([JW, Proposition 4.2]). Let $x, y \in W$. Then if ${}^p m_{y,x} \neq 0$ then $y \leq x$, $L(x) \subseteq L(y)$ and $R(x) \subseteq R(y)$.

Remark 2.10. Let $x \in W \setminus \{id\}$ and $y \in W$. If ${}^p m_{y,x} \neq 0$, then we have $\emptyset \neq L(x) \subseteq L(y)$, whence $y \neq id$.

Convention 2.11 (Longest element). We denote by w_0 the longest element of W . In type B_n (or C_n), we have

$$w_0 = s_1 s_2 s_1 s_2 [3, 1, 3] \dots [n, 1, n].$$

In type B_n and C_n , the element w_0 is central ([Da, Remark 13.1.8]). Note that

$$[i, 1, i][j, 1, j] = [j, 1, j][i, 1, i] \text{ for } 2 \leq i, j \leq n.$$

Remark 2.12. Let $x \in W$ such that ${}^p m_{x,w_0} \neq 0$. Then $L(x) = R(x) = S$, whence $x = w_0$.

Lemma 2.13 ([BB1, Exercise 2.10]). Let $w \in W$. Then

- (1) $L(ww_0) = S \setminus L(w)$ and $L(w_0w) = S \setminus w_0 L(w) w_0$
- (2) $R(ww_0) = S \setminus w_0 R(w) w_0$ and $R(w_0w) = S \setminus R(w)$.

Remark 2.14. Let $x, y \in W$. Let $s \in L(x)$. If b_y does not appear in the decomposition of $b_s {}^p b_{sx}$, then ${}^p m_{y,x} = 0$. Symmetrically, for $s \in R(x)$, if b_y does not appear in the decomposition of ${}^p b_{xs} b_s$, then ${}^p m_{y,x} = 0$. Indeed,

$$b_s {}^p b_{sx} = b_s b_{sx} + \sum_{z < sx} {}^p m_{z,sx} b_s b_z = b_x + \sum_{y < sx} \mu_{s,sx}^y b_y + \sum_{y < sx} {}^p m_{z,sx} \left(\sum_{y \leq z} \mu_{s,z}^y \right) b_y.$$

In particular, ${}^p \mu_{s,sx}^x = 1$ and if ${}^p m_{y,x} \neq 0$, then b_y appears in the decomposition of $b_s {}^p b_{sx}$.

2.2.3. *The star operation and inverse.* A second approach is to use various involutions ι on W such as the star operation or the inverse map and see how the coefficients ${}^p m_{y,x}$ and ${}^p m_{\iota(y),\iota(x)}$ are related.

Lemma 2.15 ([JW, Proposition 4.2]). *Let $x, y \in W$, then*

$${}^p m_{y,x} = {}^p m_{y^{-1},x^{-1}} \quad \text{and} \quad {}^p h_{y,x} = {}^p h_{y^{-1},x^{-1}}.$$

Definition 2.16 (Star operation). Let $s, t \in S$ such that $sts = tst$. The *left star operation* $*_{s,t}$ with respect to s and t is defined as follows. Let $w \in W$ such that $|\{s, t\} \cap L(w)| = 1$. Then $*_{s,t}w$ is the unique element in $\{sw, tw\}$ such that $|\{s, t\} \cap L(*_{s,t}w)| = 1$. Analogously, we define the *right star operation* on the right for the right descent set.

Remark 2.17. Let $s, t, w \in W$ as in the previous definition with $s \in L(w)$. In particular, there is $z \in W$ such that $w = sz$ for $z \in W$ with $l(z) = l(w) - 1$. Then the following two cases can occur.

- (1) $t \notin L(z)$. In this case, $*_{s,t}w = tw = tsz$.
- (2) $t \in L(z)$. Then there is $z' \in W$ such that $w = stz'$ and $l(z') = l(w) - 2$. Note that $s \notin L(z')$ since $sts = tst$ and $t \notin L(w)$. In this case, $*_{s,t}w = tz'$.

Lemma 2.18 ([Je1, Corollary 4.7]). *Let $s, t \in S$ such that $sts = tst$. For $x, y \in W$ with $|L(x) \cap \{s, t\}| = |L(y) \cap \{s, t\}| = 1$, we have*

$${}^p m_{y,x} = {}^p m_{*_{s,t}y, *_{s,t}x}.$$

The star operation commutes with multiplication by w_0 .

Lemma 2.19. *Let $w \in W$ and $s, t \in S$ such that $sts = tst$ and, moreover, $|\{s, t\} \cap L(w)| = 1$. Then*

$$*_{s,t}(ww_0) = *_{s,t}ww_0.$$

The same result holds for the right star operation. If, moreover, W is of type B_n or C_n , then

$$*_{s,t}(w_0w) = w_0 *_{s,t}w$$

Proof. This is a consequence of Lemma 2.13. Indeed, we may assume that $s \in L(w)$ and $t \notin L(w)$, then $s \notin L(ww_0)$ and $t \in L(ww_0)$ and we can apply the star operation. By Lemma 2.13, $s \in L(*_{s,t}ww_0)$ and $t \notin L(*_{s,t}ww_0)$. Thus, $*_{s,t}(ww_0) = *_{s,t}ww_0$. $\square$

Lastly, the star operation works well with left/right order as explained in [Je1, Corollary 4.12 and Theorem 4.13].

Lemma 2.20. *Let $x, y \in W$. Let $s, t \in S$ such that $sts = tst$ and further $|\{s, t\} \cap L(x)| = 1 = |\{s, t\} \cap L(y)|$. Then*

$$(1) \quad x \sim_L^p *_{s,t}x,$$

(2) if $x \leq_R^p y$, then ${}^{*s,t}x \leq_R^p {}^{*s,t}y$ whence, if $x \sim_R^p y$, then ${}^{*s,t}x \sim_R^p {}^{*s,t}y$
 The same result holds for the right star operation.

2.2.4. *Parabolic subgroups.* The last tool we will use is induction from parabolic subgroups of W .

Definition 2.21. Let $I \subseteq S$. We write W_I for the *parabolic subgroup* of W generated by the simple reflections in I . We denote by W^I the set of minimal length of coset representatives in W/W_I .

Lemma 2.22 ([Je1, Lemma 3.12]). Let $I \subseteq S$ and $w \in W^I$ with reduced expression $\underline{w}$ and $x \in W_I$ with a fixed reduced expression $\underline{x}$. Let $y \in W_I$, then there is a decoration and defect preserving bijection:

$$\begin{aligned} \{\text{subexpression of } \underline{x} \text{ for } y\} &\rightarrow \{\text{subexpression of } \underline{wx} \text{ for } wy\} \\ e &\mapsto \underline{w}e \end{aligned}$$

Lemma 2.23 ([Je1, Theorem 3.9]). Let $I \subseteq S$ and $x, y \in W_I$ such that $x \leq_R^p y$ in W_I if and only if for all $w \in W^I$,

$$wx \leq_R^p wy$$

in W .

Lemma 2.24 ([Je1, Corollary 3.10]). Let $I \subseteq S$ and $w \in W^I$ with reduced expression $\underline{w}$ and $x \in W_I$ with a fixed reduced expression $\underline{x}$. Let $y \leq x$. Then

$${}^p h_{wy,wx} = {}^p h_{y,x} \text{ and } h_{y,x} = h_{wy,wx}.$$

Lemma 2.25. Let $I \subseteq S$ and $w \in W^I$ with reduced expression $\underline{w}$ and $x \in W_I$ with a fixed reduced expression $\underline{x}$. Let $y \leq x$. Then

$${}^p m_{wy,wx} = {}^p m_{y,x}.$$

Proof. We proceed by induction on $l(x) - l(y)$. If $l(x) - l(y) = 0$, then $y = x$ and the result is clear. Assume now that $l(x) - l(y) = k$ and that for all $y' \in W_I$ such that $l(x) - l(y') < k$, ${}^p m_{wy',wx} = {}^p m_{y',x}$.

Recall that

$${}^p m_{y,x} = h_{y,x} - {}^p h_{y,x} - \sum_{y < z < x} {}^p m_{y,z} {}^p h_{z,x}.$$

If $z \in W$ such that $y < z < x$, then $z \in W_I$ and $wy < wz < wx$. On the other hand, if $z \in W$ with $wy < z < wx$, then $z \in wW_I$ by [BB1, Proposition 2.5.1]. Hence, there is $z' \in W_I$ such that $z = wz'$. Thus, there is a bijection

$$\begin{aligned} \{z \in W_I \mid y < z < x\} &\rightarrow \{z \in W \mid wy < z < wx\} \\ z &\mapsto wz. \end{aligned}$$

Therefore, by Lemma 2.24 and using the induction hypothesis,

$$\begin{aligned}
 {}^p m_{y,x} &= h_{y,x} - {}^p h_{y,x} - \sum_{y < z < x} {}^p m_{y,z} {}^p h_{z,x} \\
 &= h_{wy,wx} - {}^p h_{wy,wx} - \sum_{wy < wz < wx} {}^p m_{wy,wz} {}^p h_{wz,wx} \\
 &= h_{wy,wx} - {}^p h_{wy,wx} - \sum_{wy < z < wx} {}^p m_{wy,z} {}^p h_{z,wx} = {}^p m_{wy,wx}.
 \end{aligned}$$

□

3. THE SUBREGULAR CELL IN CLASSICAL TYPE

We will now describe the p -cells for the subregular two-sided 0-cell in classical types.

Convention 3.1 (The subregular cell). We write J_1 for the ordinary subregular two-sided cell, i.e. the set of non-identity elements with a unique reduced expression.

In types A_n and D_n , every element of J_1 is an oriented path:

$$J_1 = \{[a, b] \mid 1 \leq a, b \leq n\}.$$

The left cell with right descent s_i , and the right cell with left descent s_i , are

$$L_{s_i} = \{[a, i] \mid 1 \leq a \leq n\}, \quad R_{s_i} = \{[i, a] \mid 1 \leq a \leq n\}.$$

In particular,

$$L_{s_i} \cap R_{s_i} = \{s_i\}.$$

In types B_n and C_n , the elements of J_1 are of two kinds. First, there are the path elements

$$[a, b] \quad (1 \leq a, b \leq n).$$

Second, for $a, b \geq 2$, there are the elements which go down to the exceptional edge, turn around, and then go back up:

$$[a, 1, b] \quad (2 \leq a, b \leq n).$$

There is also the element $[1, 2, 1]$. Thus

$$J_1 = \{[a, b] \mid 1 \leq a, b \leq n\} \sqcup \{[a, 1, b] \mid 2 \leq a, b \leq n\} \sqcup \{[1, 2, 1]\}.$$

With the convention that L_{s_i} denotes the left cell with right descent s_i , one has

$$L_{s_1} = \{[a, 1] \mid 1 \leq a \leq n\} \sqcup \{[1, 2, 1]\},$$

and, for $i \geq 2$,

$$L_{s_i} = \{[a, i] \mid 1 \leq a \leq n\} \sqcup \{[a, 1, i] \mid 2 \leq a \leq n\}.$$

Similarly, R_{s_i} is obtained by reversing the words. Hence

$$|L_{s_1}| = n + 1, \quad |L_{s_i}| = 2n - 1 \quad (i \geq 2),$$

and

$$L_{s_1} \cap R_{s_1} = \{s_1, s_1 s_2 s_1\},$$

while, for $i \geq 2$,

$$L_{s_i} \cap R_{s_i} = \{s_i, [i, 1, i]\}.$$

Remark 3.2. The notation is chosen so that the first index records the left descent and the last index records the right descent. Thus, e.g., L_{s_i} is indexed by the right endpoint i .

3.1. The p -Kazhdan–Lusztig elements. We first describe the decomposition of ${}^p b_x$ for $x \in J_1$.

Thanks to the star operation, we inductively show that if $x \in J_1$ is of the form $[a, b]$, then ${}^p b_x = b_x$.

Lemma 3.3. *Let p be any prime number. Let $1 \leq a, b \leq n$ and $x = [a, b]$ be the shortest path. Then*

$${}^p b_x = b_x.$$

Proof. We proceed by induction on $l(x)$. If $l(x) \leq 2$, then we know that ${}^p b_x = b_x$.

Assume now that $l(x) = l > 2$ and that for all x' of the form $[a', b']$ for some $1 \leq a', b' \leq n$ with $l(x') < l$, ${}^p b_{x'} = b_{x'}$.

Let $y \leq x$ such that ${}^p m_{y,x} \neq 0$. Suppose first that $a > b$. In particular, $a \geq 3$. By Lemma 2.9, $s_a \in L(y)$. Moreover, $s_{a-1} \notin L(y)$ since y is less than x in the Bruhat order. Thus, we can apply the left star operation to x and y with respect to $\{s_a, s_{a-1}\}$ and obtain *x and *y . Note that ${}^*x = [a-1, b]$. By Lemma 2.18 and the induction hypothesis,

$${}^p m_{y,x} = {}^p m_{{}^*y, {}^*x} = \begin{cases} 1 & \text{if } {}^*y = {}^*x, \\ 0 & \text{otherwise.} \end{cases}$$

Therefore, $y = x$ and ${}^p b_x = b_x$.

Suppose now that $b > a$. By the previous case, since $x^{-1} = [b, a]$, ${}^p b_{x^{-1}} = b_{x^{-1}}$. Thus, by Lemma 2.15, ${}^p b_x = b_x$. $\square$

We can now assume that W is of type B_n or C_n and that x is of the form $[a, 1, b]$ for $1 \leq a, b \leq n$. The description of the decomposition (Theorem 3.7) will be proven by induction, the base case being when $n = 2$. We first describe the elements $z < x$ whose left or right descent sets satisfies certain properties. The following lemma provides results used towards the induction step, considering the case $n \in \{a, b\}$.

Lemma 3.4. *Assume that W is of type B_n or C_n . Let $2 \leq a, b \leq n$.*

- (1) *Let $x = [n, 1, b]$ and $z < x$ with $s_n \in L(z) \subseteq \{s_n, s_{n-1}\}$. Then, there exists $z_1 < [n-1, 1, b]$ with $z = s_n z_1$ and $l(z) = 1 + l(z_1)$. Moreover, $L(z_1) \subseteq \{s_{n-1}\}$.*

- (2) Let $x = [a, 1, n]$ and $z < x$ with $s_n \in R(z) \subseteq \{s_n, s_{n-1}\}$. Then, there exists $z_1 < [a, 1, n-1]$ with $z = z_1 s_n$ and $l(z) = 1 + l(z_1)$. Moreover, $R(z_1) \subseteq \{s_{n-1}\}$.
- (3) Let $x = [n, 1, n]$ and $z < x$ with $s_n \in L(z) \subseteq \{s_n, s_{n-1}\}$ and $s_n \in R(z) \subseteq \{s_n, s_{n-1}\}$. Then, either $z = s_n$ or there is $z_1 < [n-1, 1, n-1]$ with $z = s_n z_1 s_n$ and $l(z) = 2 + l(z_1)$. Moreover, $L(z_1) = \{s_{n-1}\}$ and $R(z_1) = \{s_{n-1}\}$.

Proof. We first prove (1), the proof of (2) being entirely symmetric.

By [BB1, Proposition 2.5.1], there is $z_1 < [n-1, 1, b]$ and $z_0 \in W_{\{s_n\}}$ such that $z = z_0 z_1$ and $l(z) = l(z_0) + l(z_1)$. Assume for a contradiction that z_0 is the identity, i.e. $z < [n-1, 1, b]$. Since $s_n \in L(z)$, we must have $z \in W_{\{s_1, \dots, s_{n-2}\} s_n}$ (and $b = n$). If $z \neq s_n$, there is $1 \leq i \leq n-2$, such that $s_i \in L(z)$, which contradicts $L(z) \subseteq \{s_n, s_{n-1}\}$. Thus, $z = s_n z_1$ with $z_1 < [n-1, 1, b]$ and $l(z) = 1 + l(z_1)$. Moreover, $L(z_1) \subseteq \{s_{n-1}\}$ since $L(z) = \{s_n, s_{n-1}\}$.

We now check (3), which is a corollary of (1) and (2). Assume that $z \neq s_n$. By (1), there exists $id \neq z_2 < [n-1, 1, n]$ with $z = s_n z_2$ with $l(z) = 1 + l(z_2) > 1$. Moreover, $L(z_2) = \{s_{n-1}\}$. Observe that $R(z_2) \subseteq \{s_n, s_{n-1}\}$. If $s_n \notin R(z_2)$, then $z_2 < [n-1, 1, n-1]$. Now that, since $s_n \in R(z)$, we must in fact have $z_2 \in W_{\{s_1, \dots, s_{n-2}\}}$. Since $z_2 \neq id$, this contradicts $R(z_2) \subseteq \{s_n, s_{n-1}\}$. Thus, $s_n \in R(z_2)$ and we may apply (2) to $z_2 < [n-1, 1, n]$. There exists $z_1 < [n-1, 1, n-1]$ with $z_2 = z_1 s_n$ and $l(z_2) = 1 + l(z_1)$. Since $L(z_2) = \{s_{n-1}\}$, z_1 is not the identity. Moreover, $R(z_1) = \{s_{n-1}\}$. Therefore, $z = s_n z_1 s_n$ for $z_1 < [n-1, 1, n-1]$ and $L(z_1) = R(z_1) = \{s_{n-1}\}$. $\square$

Lemma 3.5. Assume that W is of type B_n or C_n . Let $2 \leq a, b \leq n$ and $x = [a, 1, b]$. Let $z \leq x$ such that $L(z) = \{s_a\}$ and $R(z) = \{s_b\}$. Then

- either $z = [a, b]$
- or $z = [a, 1, b]$.

Proof. If $n = 2$, then it is clear as $x = s_2 s_1 s_2$. Assume now that the result holds for B_k with $k < n$. In particular, if $a, b < n$, then $z \in \{[a, b], [a, 1, b]\}$. Thus, we may assume that $n \in \{a, b\}$.

Suppose first that $a = n$. If $z = s_n$, then $b = n$ and $z = [n, n]$. Thus, we may assume that $z \neq s_n$. By Lemma 3.4 and $R(z_1) = \{s_b\}$, there exists $z_1 < [n-1, 1, b]$ with $z = s_n z_1$ with $l(z) = 1 + l(z_1)$. Moreover, $L(z_1) \subseteq \{s_{n-1}\}$ and $R(z_1) = \{s_b\}$.

If $b < n$, then, by the induction hypothesis, $z_1 \in \{[n-1, b], [n-1, 1, b]\}$, whence $z \in \{[n, b], [n, 1, b]\}$.

If $b = n$, then by Lemma 3.4, there is $z_1 < [n-1, 1, n-1]$ such that $z = s_n z_1 s_n$ with $l(z) = l(z_1) + 2$. Moreover, $L(z_1) = \{s_{n-1}\} = R(z_1)$. By the induction hypothesis, $z_1 \in \{s_{n-1}, [n-1, 1, n-1]\}$. Since $L(z) = \{s_n\}$, $z_1 \neq s_{n-1}$, so $z_1 = [n-1, 1, n-1]$ and $z = [n, 1, n]$.

The case where $a < n$ and $b = n$ is similar. $\square$

In this lemma, we compute the coefficient of $b_{[n,n-1,n]}$ in ${}^p b_x$ for $x = [n, 1, n]$ in type C_n using the graded rank of the local intersection form, (see 2.2.1).

Lemma 3.6. *Assume that W is of type C_n and $n \geq 3$. Let $x = [n, 1, n]$ and $y = s_n s_{n-1} s_n$. Then*

$${}^p m_{y,x} = \begin{cases} 1, & p = 2 \\ 0 & \text{otherwise.} \end{cases}$$

Proof. Recall that

$${}^p m_{y,x} = h_{y,x} - {}^p h_{y,x} - \sum_{y < z < x} {}^p m_{y,z} {}^p h_{z,x}.$$

We first compute $h_{y,x}$ and ${}^p h_{y,x}$ as the graded rank of the local intersection form, following [EMTW, Section 27.3]. The subexpressions of $[n, 1, n]$ expressing y are either of the form

- (1) $11e0\tilde{e}01$ for some subexpression e of $[n-2, 2]$ with $\tilde{e}$ the symmetric of e in $[2, n-2]$,
- (2) or $10e0\tilde{e}11$ for some subexpression e of $[n-2, 2]$ with $\tilde{e}$ the symmetric of e in $[2, n-2]$.

Let n_e be the number of 0 in e . In the first case, the subexpression has defect $n_e + 1 + n_e - 1$. In the second case, it has defect $1 + n_e + 1 + n_e$. Thus, all subexpressions have non-negative defects. Therefore, the d -graded piece of the local intersection form is zero unless $d = 0$. In that case, it is a single entry matrix of the form $(-2)^{n-2}$. Thus, $h_{y,x} = 1$ and

$${}^p h_{y,x} = \begin{cases} 0, & p = 2 \\ 1 & \text{otherwise.} \end{cases}$$

In particular, ${}^p m_{y,x} = 0$ if $p \neq 2$.

We now study the sum $\sum_{y < z < x} {}^p m_{y,z} {}^p h_{z,x}$ and show that there is no $y < z < x$ such that ${}^p m_{y,z} \neq 0$. Let $y < z \leq x$ such that ${}^p m_{y,z} \neq 0$. If ${}^p m_{y,z} \neq 0$, then $L(z) \subseteq L(y) = \{s_{n-1}, s_n\}$ and $R(z) \subseteq R(y) = \{s_{n-1}, s_n\}$ by Lemma 2.9. Note that since $y < z < x$, we must have $s_n \in L(z)$ and $s_n \in R(z)$. By Lemma 3.4, since $z \leq x$, there is $z_1 < [n-1, 1, n-1]$ with $l(z) = 2 + l(z_1)$ and $L(z_1) = \{s_{n-1}\} = R(z_1)$. We conclude that $z_1 \in \{s_{n-1}, [n-1, 1, n-1]\}$, by Lemma 3.5. Thus, $z \in \{s_n s_{n-1} s_n, [n, 1, n]\} = \{x, y\}$ and since $z > y$, $z = x$. Therefore, the sum $\sum_{y < z < x} {}^p m_{y,z} {}^p h_{z,x} = 0$ and ${}^p m_{y,x} = h_{y,x} - {}^p h_{y,x} = 1$ if $p = 2$. $\square$

Theorem 3.7 (p -canonical basis for the subregular cell). *Let $x \in J_1$.*

- (a) *In types A_n and D_n , for any prime p , one has*

$${}^p b_x = b_x.$$

- (b) In type B_n , if $p \neq 2$, then ${}^p b_x = b_x$. If $p = 2$, the only change inside the ordinary subregular cell is:

$${}^p b_{[1,2,1]} = b_{[1,2,1]} + b_1.$$

All other elements $x \in J_1$ satisfy ${}^p b_x = b_x$.

- (c) In type C_n , if $p \neq 2$, then ${}^p b_x = b_x$. If $p = 2$, the only elements of J_1 which change are of the form $[a, 1, b]$ with $a, b \geq 2$. For these one has

$${}^p b_{[a,1,b]} = \sum_{k=1}^{\min(a,b)} b_{[a,k,b]}.$$

All other elements $x \in J_1$ satisfy ${}^p b_x = b_x$.

Proof. If W is of type A_n or D_n , this is a direct consequence of Lemma 3.3.

We can now assume that W is of type B_n or C_n . We proceed by induction on n . The case $n = 2$ can be found in [JW, Example 5.1]. Assume now that $n \geq 3$ and that the result holds for B_{n-1} and C_{n-1} . Let $x \in W$.

Thanks to Lemma 3.3, if $x = [a, b]$ for some $1 \leq a, b \leq n$, then

$${}^p b_x = b_x.$$

By the induction hypothesis, if $x = [a, 1, b]$ for $2 \leq a, b < n$, then

- in type B_n , as well as in type C_n for $p \neq 2$,

$${}^p b_x = b_x,$$

- while in type C_n for $p = 2$,

$${}^p b_x = \sum_{k=1}^{\min(a,b)} b_{[a,k,b]}.$$

If $x = [1, 2, 1]$, by the induction hypothesis,

- in type B_n for $p = 2$,

$${}^p b_{s_1 s_2 s_1} = b_{s_1 s_2 s_1} + b_1,$$

- while, in type C_n , as well as in type B_n for $p \neq 2$,

$${}^p b_{s_1 s_2 s_1} = b_{s_1 s_2 s_1}.$$

We may thus assume that $x = [a, 1, b]$ for some $2 \leq a, b \leq n$ with $n \in \{a, b\}$. Let $y \leq x$ such that ${}^p m_{y,x} \neq 0$.

Case 1: Suppose first that $x = [a, 1, n]$ for some $2 \leq a < n$. By Lemma 2.9, $s_n \in R(y)$. Moreover, $s_{n-1} \notin R(y)$ since $y \leq [a, 1, n]$ and $a < n$. We apply the right star operation to x and y with respect to $\{s_n, s_{n-1}\}$ to obtain x^* and y^* . Note that $x^* = [a, 1, n-1]$. By Lemma 2.18 and the induction hypothesis, we obtain

- in type B_n , as well as in type C_n for $p \neq 2$,

$${}^p m_{y,x} = {}^p m_{y^*,x^*} = \begin{cases} 1 & \text{if } y^* = x^*, \\ 0 & \text{otherwise,} \end{cases}$$

- while, in type C_n for $p = 2$,

$${}^p m_{y,x} = {}^p m_{y^*,x^*} = \begin{cases} 1 & \text{if } y^* = [a, k, n-1] \text{ for some } 1 \leq k \leq a, \\ 0 & \text{otherwise.} \end{cases}$$

Observing that if $y^* = [a, k, n-1]$ for some $1 \leq k \leq a$, then $y = [a, k, n]$, we deduce that

- in type B_n , as well as in type C_n for $p \neq 2$,

$${}^p b_x = b_x,$$

- while, in type C_n for $p = 2$,

$${}^p b_{[a,1,n]} = \sum_{k=1}^{\min(a,n)} b_{[a,k,n]}.$$

Case 2: Assume that $x = [n, 1, b]$ for some $2 \leq b < n$. Note that $x^{-1} = [b, 1, n]$ corresponds to **Case 1**. Applying Lemma 2.15 allows us to conclude that,

- in type B_n , as well as in type C_n for $p \neq 2$,

$${}^p b_x = b_x$$

- while, in type C_n for $p = 2$,

$${}^p b_x = \sum_{y \in W} {}^p m_{y^{-1},x^{-1}} b_y = \sum_{k=1}^b b_{[b,k,n]^{-1}} = \sum_{k=1}^b b_{[n,k,b]}.$$

We may now finally assume that $x = [n, 1, n]$.

Case 3a: Assume that $x = [n, 1, n]$ and that $s_{n-1} \notin L(y)$. We apply the left star operation to x and y with respect to $\{s_n, s_{n-1}\}$ to obtain *x and *y . Since ${}^*x = [n-1, 1, n]$, Lemma 2.18 and **Case 1** imply that:

- in type B_n , as well as in type C_n for $p \neq 2$,

$${}^p m_{y,x} = {}^p m_{{}^*y,{}^*x} = \begin{cases} 1 & \text{if } {}^*y = {}^*x \iff y = x, \\ 0 & \text{otherwise,} \end{cases}$$

- while, in type C_n for $p = 2$,

$${}^p m_{y,x} = {}^p m_{{}^*y,{}^*x} = \begin{cases} 1 & \text{if for some } 1 \leq k \leq n-1, \quad {}^*y = [n-1, k, n], \\ 0 & \text{otherwise.} \end{cases}$$

We observe that if $*y = [n-1, k, n]$, then $y = [n, k, n]$ unless $k = n-1$, in which case $y = s_n$.

Case 3b: Assume that W is either of type B_n or of type C_n with $p \neq 2$, and that $x = [n, 1, n]$. We show that if $y < x$ with ${}^p m_{y,x} \neq 0$ then $s_{n-1} \notin L(y)$. Since ${}^p m_{y,x} \neq 0$, we see that b_y appears as a summand in $b_{s_n} {}^p b_{[n-1,1,n]}$ by Remark 2.14. Now by **Case 1**, we know ${}^p b_{[n-1,1,n]}$. We compute

$$\begin{aligned} b_{s_n} {}^p b_{[n-1,1,n]} &= b_{s_n} b_{[n-1,1,n]} \\ &= b_{[n,1,n]} + \sum_{z < [n-1,1,n]} \mu_{s_n, [n-1,1,n]}^z b_z. \end{aligned}$$

In particular, since $y < x$, we must have $y < [n-1, 1, n]$. Since $s_n \in R(y)$, there is $y_1 < [n-1, 1, n-1]$ such that $y = y_1 s_n$ with $l(y) = l(y_1) + 1$. Now, since $s_n \in L(y)$, we must have $y_1 \in W_{\{s_1, \dots, s_{n-2}\}}$ and in particular, $s_{n-1} \notin L(y)$.

Therefore, if W is of type B_n or of C_n with $p \neq 2$, then by **Case 3a**,

$${}^p b_{[n,1,n]} = b_{[n,1,n]}.$$

Case 3c: Lastly, assume that W is of type C_n , $p = 2$ and $x = [n, 1, n]$. Using **Case 1** we have

$$\begin{aligned} b_{s_n} {}^p b_{[n-1,1,n]} &= \sum_{k=1}^{n-1} b_{s_n} b_{[n-1,k,n]} \\ &= \sum_{k=1}^{n-1} b_{[n,k,n]} + \sum_{k=1}^{n-1} \sum_{z < [n-1,k,n]} \mu_{s_n, [n-1,k,n]}^z b_z \end{aligned}$$

By the same reasoning as for **Case 3b**, we conclude that, if ${}^p m_{y,x} \neq 0$, then either $y \in W_{\{s_1, \dots, s_{n-2}\}} s_n$ (in particular $s_{n-1} \notin L(y)$) or $y = [n, k, n]$ for some $1 \leq k \leq n$. By **Case 3a**, we know that, if $s_{n-1} \notin L(y)$, then ${}^p m_{y,x} = 1$ if and only if $y = [n, k, n]$ for some $1 \leq k \leq n$ (with $k \neq n-1$). If $s_{n-1} \in L(y)$, then $y = s_n s_{n-1} s_n$ and ${}^p m_{y,x} = 1$ by Lemma 3.6. Therefore,

$${}^p b_{[n,1,n]} = \sum_{k=1}^n b_{[n,k,n]}.$$

This concludes the description of the elements ${}^p b_x$ for $x \in J_1$. $\square$

3.2. The p -cells coming from the subregular cell. We now consider how the cell J_1 distributes in characteristic p . We start by showing that J_1 decomposes into a union of p -cells.

The first step is to verify that for $v \in J_1$, the canonical basis element b_v does not appear as a coefficient of any ${}^p b_w$ for $w \notin J_1$.

Lemma 3.8. *Let $w \in W \setminus \{id\}$ and $v \in J_1$. Then*

$${}^p m_{v,w} \neq 0 \implies w \in J_1.$$

Proof. We first observe that for $w \in W \setminus \{id\}$, we always have $w \geq_J^0 v$. Indeed, let $s \in L(w)$. Then, $\mu_{s,sw}^w = 1$, hence $s \leq_R^0 w$. Thus, $s \leq_J^0 w$ and since $s \in J_1$, also $v \leq_J^0 w$.

Now let $w \in W \setminus \{id\}$ and $v \in J_1$. We show that, if ${}^p m_{v,w} \neq 0$, then $w \in J_1$. We proceed by induction on $l(w)$. If $l(w) = 1$, then ${}^p b_w = b_w$ and $v = w \in J_1$. Now assume that $l(w) > 1$ and ${}^p m_{v,w} \neq 0$. By Lemma 2.9, $L(w) \subseteq L(v)$. Let $s \in S$ such that $L(v) = \{s\}$. Then b_v appears as summand of $b_s {}^p b_{sw}$ (see Remark 2.14). In other words, either $v = w$ and $w \in J_1$ or there is $v' \in W$ such that ${}^p m_{v',sw} \neq 0$ and $v \leq_L^0 v'$. If $v' = id$, then $w = s$ and $w \in J_1$. Thus, we may assume that $v' \neq id$, and so $v \leq_J^0 v'$. Therefore, $v \sim_J^0 v'$, whence $v' \in J_1$. By induction hypothesis, we then have $sw \in J_1$. Knowing that $L(w) = \{s\}$ and using the description of the subregular cell J_1 (Convention 3.1), we deduce that $w \in J_1$. $\square$

Next we will show that J_1 decomposes as a union of p -cells.

Lemma 3.9 (Union of p -cells). *Let $x \in J_1$ and $y \in W \setminus \{id\}$. Then for $X \in \{L, R, J\}$, if $x \geq_X^p y$, then $y \in J_1$. In particular, for any $y \in W \setminus \{id\}$, either $y \in J_1$ or $y \not\sim_J^p x$.*

Proof. Consider the case $X = R$, the case $X = L$ being analogous. Since $x \geq_R^p y$, there is $z \in W$ such that ${}^p \mu_{z,y}^x \neq 0$. Thus, there is $v, w \in W$ such that ${}^p m_{v,z} {}^p m_{w,y} \mu_{v,w}^x \neq 0$ by (2.2). Since $\mu_{v,w}^x \neq 0$, we have $x \geq_R^0 w$, hence $x \sim_J^0 w$. Moreover, since ${}^p m_{w,y} \neq 0$, we must have $y \in J_1$ by the previous Lemma.

Suppose that $x \geq_J^p y$. Then there is $w \in W$ such that $x \geq_R^p w$ and $w \geq_L^p y$. Thus, $w \in J_1$ and $y \in J_1$.

We conclude by observing that if $y \sim_J^p x$, then $x \geq_J^p y$, whence $y \in J_1$. $\square$

We will now compute the p -cells in low rank, which will serve as a base for induction.

Lemma 3.10. *Assume $p = 2$ and consider Weyl groups of low rank. Then the subregular cell J_1 splits into p -cells as follows.*

- (a) *If W is of type A_2 , then J_1 is a two-sided p -cell.*
- (b) *If W is of type B_2 , then J_1 decomposes into two two-sided p -cells: $\{s_1\}$ and $J_1 \setminus \{s_1\}$.*
- (c) *If W is of type C_2 , then J_1 decomposes into two two-sided p -cells: $\{s_2\}$ and $J_1 \setminus \{s_2\}$.*

Proof. This is an explicit computation in the p -canonical basis. In type A_2 , the p -canonical basis is the same as ordinary canonical basis. In the notation of Lemma 3.7, in type B_2 , the only change is

$${}^p b_{[1,2,1]} = b_{[1,2,1]} + b_1,$$

while in type C_2 , the only change is

$${}^p b_{[2,1,2]} = b_{[2,1,2]} + b_2.$$

The resulting p -cell preorder is the one stated above. $\square$

The following theorem describes the subregular p -cells in classical types. Note that in type A_n , Jensen showed much more generally that p -cells and 0-cells coincide, see [Je1, Theorem 4.33]. We include this case for completeness and also give the proof with our methods to facilitate understanding of the other types. The notation $a_{r,s}$ denotes an $r \times s$ block all of whose entries are a . Thus $2_{n-1,n-1}$ is the $(n-1) \times (n-1)$ matrix where each entry is 2.

Theorem 3.11 (The subregular p -cells). *If $p \neq 2$ or if W is of type A_n, D_n , the subregular 0-cell $\mathcal{J}_1$ is a two-sided p -cell. It contains the longest element of a parabolic subgroup.*

In types B_n and C_n , if $p = 2$, the ordinary subregular cell J_1 is the union of two two-sided p -cells. Order the left and right cells by

$$(s_1 \mid s_2, \dots, s_n),$$

Then:

- (a) *In type B_n , one p -cell is strongly regular of size 1 and contains s_1 , and the other has shape*

$$\frac{1_{1,1} \mid 1_{1,n-1}}{1_{n-1,1} \mid 2_{n-1,n-1}}.$$

It also contains the longest element of a parabolic subgroup.

- (b) *In type C_n , one p -cell is strongly regular of size $(n-1) \times (n-1)$, and the other has shape*

$$\frac{2_{1,1} \mid 1_{1,n-1}}{1_{n-1,1} \mid 1_{n-1,n-1}}.$$

The strongly regular p -cell contains the elements of the form $[a, b]$ for $2 \leq a, b \leq n$. Both p -cells contain the longest element of a parabolic subgroup.

Proof. We use the known relations for Weyl groups of rank 2 and 3 and propagate them using star operations, concluding by a counting argument.

Type A_n . Assume that W is of type A_n .

We first show that the left 0-cell containing 1 does not split, i.e. for all $1 \leq a \leq n$,

$$[a, 1] \sim_L^p s_1.$$

If $a = 2$, then it is Lemma 3.10 applied to $W_{\{s_1, s_2\}}$. By transitivity of the left cell order, it is enough to show that, for any $2 < a \leq n$, we have $[a, 1] \sim_L^p [a-1, 1]$.

By applying right star operations successively, each time using Lemma 2.20 and [Je1, Theorem 3.9], we have

$$[a, 1] \sim_L^p [a - 1, 1] \iff [a, a - 1] \sim_L^p [a, a],$$

and the right-hand side of the equivalence holds by Lemma 3.10 applied to $W_{\{s_a, s_{a-1}\}}$.

Thanks again to Lemma 2.20, we thus conclude that each left 0-cell is left unchanged. Thus, the whole subregular cell J_1 is a two-sided p -cell.

Type D_n . Assume that W is of type D_n for $n \geq 4$. Restricting to $W_{\{s_2, s_3, \dots, s_n\}}$, we see that all shortest paths of the form $[a, b]$ for $2 \leq a, b \leq n$ lie in the same two-sided p -cell. On the other hand, restricting to $W_{\{s_1, s_3, \dots, s_n\}}$, we see that for all $3 \leq a \leq n$

$$s_1 \sim_L^p [a, 1] \sim_R^p s_a \sim_L^p [1, a].$$

Thus, the whole subregular cell J_1 is a two-sided p -cell.

Types B_n and C_n . Assume now that W is of type B_n or C_n . By restricting to $W_{\{s_2, \dots, s_n\}}$, we observe that all the shortest paths of the form $[a, b]$ for $2 \leq a, b \leq n$ lie in the same two-sided p -cell. Moreover, by [Je1, Theorem 3.9], if $x \leq_R^p y \in W_I$, then for any $w \in W^I$, $wx \leq_R^p wy$. In particular, for $2 \leq a, b, c \leq n$, since $[2, b] \sim_R^p [2, c]$, we must have $[a, 1, b] \sim_R^p [a, 1, c]$ and $[b, 1, a] = [a, 1, b]^{-1} \sim_L^p [c, 1, a]$. Therefore, all the elements of the form $[a, 1, b]$ for $2 \leq a, b \leq n$ lie in the same two-sided p -cell.

Using Lemma 2.20 again, we observe the following, for $2 \leq a, b \leq n$:

- $[2, 2] \sim_L^p [2, 1, 2] \iff [2, b] \sim_L^p [2, 1, b],$
- $[2, 2] \sim_L^p [2, 1, 2] \iff [a, 2] \sim_L^p [a, 1, 2],$
- $[2, 1, 2] \sim_J^p [2, 1] \iff [a, 1, 2] \sim_J^p [a, 1]$
- and $[2, 1, 2] \sim_J^p [1, 2] \iff [2, 1, b] \sim_J^p [1, b].$

Thus, by Lemma 3.10, in type B_n , when $p = 2$, the two-sided p -cell containing s_2 contains every element of J_1 except $\{s_1\}$. In type C_n , when $p = 2$, the two-sided p -cell containing s_2 contains exactly the shortest paths of the form $[a, b]$ for $2 \leq a, b \leq n$. Moreover, the two-sided p -cell containing $[2, 1, 2]$ consists exactly of the remaining elements of J_1 . $\square$

4. THE SUBMAXIMAL CELL IN CLASSICAL TYPE

Following the strategy of the previous section, we will now describe the p -cells in the submaximal cell.

Convention 4.1 (The submaximal cell). Let W be of classical type. We write

$$\overline{J_1} = w_0 J_1$$

for the **submaximal** two-sided cell, i.e. the cell closest to w_0 . In general, for an element $w \in W$, we write

$$\overline{w} := w_0 w.$$

The shape of the ordinary cell $\overline{J_1}$ is the same as the shape of J_1 . Ordered by

$$(s_1 \mid s_2, \dots, s_n),$$

i.e. the exceptional endpoint first and the simply-laced tail second, it is

$$\begin{array}{cc} 2_{1,1} & 1_{1,n-1} \\ 1_{n-1,1} & 2_{n-1,n-1}. \end{array}$$

Remark 4.2. The notation is chosen so that the formulae are precisely the w_0 -translates of the notation in the subregular cell. In particular, we do not use any new combinatorics for the ordinary cell shape.

4.1. The p -Kazhdan–Lusztig elements. We first consider how the p -canonical basis elements in $\overline{J_1}$ could possibly decompose.

The first step is to reduce the possibilities of the b_y appearing as coefficients in the ${}^p b_x$ for $x \in \overline{J_1}$. In particular, we show that $y \in \overline{J_1}$ unless W is of type B_n or C_n and x is of the form $w_0 s_1$ or $w_0 s_1 s_2 s_1$. To do so, we extensively use the fact that the left and right descent of y are completely determined by the one of x . We then proceed by induction repeatedly using the star operation.

Lemma 4.3. *Let $y \in W$ such that ${}^p m_{y,x} \neq 0$ for some $x \in \overline{J_1}$. If $y \notin \overline{J_1}$, then W is of type B_n or C_n and $y = \overline{v}$ where v is of the form $s_1 s_2 s_1 [3, 1] \dots [k, 1]$ for $3 \leq k \leq n$.*

Proof. Let $y \in W$ and $x \in \overline{J_1}$ such that ${}^p m_{y,x} \neq 0$. Since $y \leq x$ in the Bruhat order, we must have $y \neq w_0$.

By definition of $\overline{J_1}$ and Lemma 2.9, we have $L(x) = L(y)$ and $R(x) = R(y)$ with both sets of size $n - 1$. Let $1 \leq a, b \leq n$ such that $s_a \notin L(y)$ and $s_b \notin R(y)$. In other words, $L(\overline{y}) = \{w_0 s_a w_0\}$ and $R(\overline{y}) = \{s_b\}$. There exists $z \in W$ such that $\overline{y} = z s_b$ and $l(\overline{y}) = l(z) + 1$. Moreover, if $s \in R(z)$, then $s \in R(\overline{y})$ unless $ss_b \neq s_b s$, whence $R(z) \subseteq \{s_{b+1}, s_{b-1}\}$ (if $1 \leq b+1, b-1 \leq n$). We will repeatedly use the following fact, a consequence of Lemma 2.20: *if we apply a (left/right) star operation to x then we obtain a new element x^* which also belongs to $\overline{J_1}$.*

Case 0 If $R(z) = \emptyset$, then $y = \overline{s_b} \in \overline{J_1}$.

Case 1 Assume that $b+1 \in R(z)$. If W is of type B_n or C_n , assume furthermore that $b \geq 2$. We use induction on the length of $\overline{y}$ to show that $y \in \overline{J_1}$.

If $l(\overline{y}) = 2$, then $\overline{y} = s_{b+1} s_b \in J_1$ and $y \in \overline{J_1}$.

Now assume that $l(\overline{y}) = l > 2$. Assume the following induction hypothesis: let $y' \in W$ with $l(\overline{y'}) < l$ such that ${}^p m_{y',x'} \neq 0$ for some $x' \in \overline{J_1}$. Let $1 \leq c \leq n$ ($c \geq 2$ if W is of type B_n or C_n) with $R(x') = S \setminus \{s_c\}$. If $y' = z' s_c$ for some $z' \in W$ with $s_{c+1} \in R(z')$ and $l(y') = l(z') + 1$, then $y' \in \overline{J_1}$.

We apply the right star operation to y and x with respect to $\{s_b, s_{b+1}\}$ (or $\{s_1, s_3\}$ if $b = 1$ and W is of type D) to obtain y^* and $x^* \in \overline{J_1}$. By definition of the star operation and since $x^* \in \overline{J_1}$, we have $R(x^*) = S \setminus \{s_{b+1}\}$. By Lemma

2.18, ${}^p m_{y^*, x^*} = {}^p m_{y, x} \neq 0$. Therefore $R(y^*) = S \setminus \{s_{b+1}\}$ and $R(\overline{y^*}) = \{s_{b+1}\}$. Hence, there is $z' \in W$ such that $\overline{y^*} = z' s_{b+1}$, $l(\overline{y^*}) = l(z') + 1$ and $R(z') \subseteq \{s_{b+2}, s_b\}$. Since $l > 2$, $R(z') \neq \emptyset$. On the other hand, $s_b \notin R(z')$ since otherwise $s_{b+1} \in R(y)$. Thus, $s_{b+2} \in R(z')$. By the induction hypothesis, $y^* = \bar{z} \in \overline{J_1}$ whence $y \in \overline{J_1}$ by Lemma 2.20.

Case 2 Suppose now that $s_{b-1} \in R(z)$. In type B_n and C_n , assume furthermore that $b \geq 2$. We check that $y \in \overline{J_1}$.

Step 1 We show that either $y \in \overline{J_1}$ or W is of type B_n or C_n and $\overline{y} = w[1, b]$ for $w \in W$ with $l(\overline{y}) = l([1, b]) + l(w)$.

If W is of type B_n or C_n and $b = 2$, then since $s_1 \in R(z)$, $\overline{y} = w[1, 2]$ for $w \in W$ with $l(\overline{y}) = 2 + l(w)$. Thus, if W is of type B_n or C_n , we may assume that $b \geq 3$.

Let us now proceed with an inductive proof. If $l(\overline{y}) = 2$, then $\overline{y} = s_{b-1}s_b$ and $y \in \overline{J_1}$.

Now assume that $l(\overline{y}) = l > 2$. We apply the right star operation to y and x with respect to $\{s_b, s_{b-1}\}$ to get y^* and $x^* \in \overline{J_1}$. Note that $\overline{y^*} = z$ and $y^* = w_0 z$ by Lemma 2.19. Thus, $l(\overline{y^*}) = l - 1 < l$. By Lemma 2.18, ${}^p m_{y^*, x^*} = {}^p m_{y, x} \neq 0$. In particular, $R(\overline{y^*}) = \{s_{b-1}\}$ and there is $z' \in W$ such that $\overline{y^*} = z' s_{b-1}$ and $l(\overline{y^*}) = l(z') + 1 = l - 1$. As before, $R(z') \subseteq \{s_b, s_{b-2}\}$ and $R(z') \neq \emptyset$ since $l(z') = l(\overline{y}) - 2 \geq 1$. If $s_b \in R(z')$, then $s_{b-1} \in R(\overline{y})$, a contradiction. Thus, $R(z') = \{s_{b-2}\}$. If $b - 1 = 2$, then $\overline{y} = w s_1 s_2 s_3$ for some $w \in W$ with $l(\overline{y}) = 3 + l(w)$. If $b - 1 > 2$, then by induction hypothesis, either $y^* \in \overline{J_1}$, whence $y \in \overline{J_1}$ or $\overline{y^*} = w[1, b - 1]$ for $w \in W$ with $l(\overline{y^*}) = l([1, b - 1]) + l(w)$, whence $\overline{y} = w[1, b]$.

Step 2 Assume that W is of type B_n or C_n and that $\overline{y} = w[1, b]$ for some $w \in W$ with $l(\overline{y}) = l(w) + l([1, b])$ for some $b > 1$. We show that $y \in \overline{J_1}$.

If $w = id$, then $\overline{y} = [1, b]$ and $y \in \overline{J_1}$. So we may assume that $w \neq id$. We will prove that $\overline{y} = [a, 1, b]$ for some $1 \leq a \leq n$.

By repeatedly applying star operations as in **Step 1**, we deduce that there exists $x' \in \overline{J_1}$ such that ${}^p m_{\overline{w s_1 s_2}, x'} \neq 0$. Let $y' = \overline{w s_1 s_2}$. If $s_3 \in R(w)$, then by **Case 1**, $y' \in \overline{J_1}$, whence w is of the form $[a, 2]$ for some $2 \leq a \leq n$, a contradiction. Thus, $s_3 \notin R(w)$.

We will show that for any $l(\overline{y'}) - 2 \geq j \geq 1$, there is $w_j \in W$ such that $\overline{y'} = w_j[j, 1, 2]$ (where $[1, 1, 2]$ denotes $[1, 2]$) with $l(\overline{y'}) = j + l(w_j) + 1$ and $R(w_j) \subseteq \{s_{j+1}\}$.

If $j = 1$, we set $w_1 = w$. We need to check that $R(w) = \{s_2\}$. By assumption, $R(w) \neq \emptyset$. Since ${}^p m_{y', x'} \neq 0$ and $x' \in \overline{J_1}$, we know that $|R(y')| = n - 1$ and $R(\overline{y'}) = \{s_2\}$. Thus $R(w) \subseteq \{s_3, s_2\}$, and since we have shown that $s_3 \notin R(w)$, we conclude that $R(w) = \{s_2\}$.

Therefore, there exists $w_2 \in W$ such that $w = w_2 s_2$ and $l(w) = l(w_2) + 1$. Since $R(\overline{y'}) = \{s_2\}$, we must have $R(w_2) \subseteq \{s_3\}$. Assume now that there is $w_k \in W$

such that $\overline{y'} = w_k[k, 1, 2]$ and $R(w_k) \subseteq \{s_{k+1}\}$ for $2 \leq k < l(\overline{y'}) - 2$. Since $k < l(\overline{y'}) - 2$, we have $w_k \neq id$ and $R(w_k) = \{s_{k+1}\}$. Thus, there is $w_{k+1} \in W$ such that $w_k = w_{k+1}s_{k+1}$ and $l(w_k) = l(w_{k+1}) + 1$. Since $R(w_k) = \{s_{k+1}\}$, we see that $R(w_{k+1}) \subseteq \{s_k, s_{k+2}\}$. If $R(w_{k+1}) = \emptyset$, then $\overline{y'} = [k+1, 1, 2]$. If $s_k \in R(w_{k+1})$, then we can write $\overline{y'} = w's_k s_{k+1} s_k[k-1, 1, 2] = w's_{k+1}[k, 1, 2]s_{k+1}$ for some $w' \in W$ with $l(\overline{y'}) = l(w') + k + 3$, contradicting $R(\overline{y'}) = \{s_2\}$. Thus $R(w_{k+1}) \subseteq \{s_{k+2}\}$. We have shown that $\overline{ws_1s_2} = [a, 1, 2] \in \overline{J_1}$. In particular $w = [a, 2]$, whence $\overline{y} = [a, 1, b] \in J_1$ and $y \in \overline{J_1}$.

Case 3: Assume that W is of type B_n or C_n and that $b = 1$. In this case, $L(\overline{y}) = \{w_0s_aw_0\} = \{s_a\}$. If $a \neq 1$, a similar analysis on the left implies that $y \in \overline{J_1}$. Thus, we may now assume that $b = a = 1$.

Observe first that if $\overline{y} = s_1s_2s_1[3, 1] \dots [k, 1]$ then $L(\overline{y}) = R(\overline{y}) = \{s_1\}$. We show by a counting argument that there is no other $w \in W$ which satisfies $L(w) = R(w) = \{s_1\}$. We set

$$X := \{w \in W \mid L(w) = R(w) \subseteq \{s_1\}\}.$$

The set X is in bijection with the set of double cosets $W_I \backslash W / W_I$ for $I = S \setminus \{s_1\}$, see [BB1, Exercise 2.15]. The number of these double cosets is equal to $\langle \text{Ind}_{W_I}^W 1_{W_I}, \text{Ind}_{W_I}^W 1_{W_I} \rangle$ [St, Exercise 7.77a]. Since $\text{Ind}_{W_I}^W 1_{W_I}$ decomposes into a sum of $n+1$ irreducible representations, each appearing with coefficient 1 (see [GP, Proposition 6.15]), we conclude that $|X| = n+1$. Note that the identity belongs to X and is the only element with empty left or right descent set. Thus, there are exactly n elements with left and right descent sets equal to $\{s_1\}$, the ones that we have enumerated above. $\square$

We are now ready to study the decomposition of the ${}^p b_x$ for $x \in \overline{J_1}$.

Theorem 4.4 (*p*-canonical basis for the submaximal cell). *Let $x \in \overline{J_1}$.*

- (1) *In types A_n and D_n , we have $b_x = {}^p b_x$ for any prime p .*
- (2) *In type B_n , if $p \neq 2$, then ${}^p b_x = b_x$. If $p = 2$, then ${}^p b_x = b_x$ unless $x = w_0[i, j]$ for $i, j \geq 2$. In this case,*

$${}^p b_{w_0[i, j]} = b_{w_0[i, j]} + b_{w_0[i, 1, j]}.$$

- (3) *In type C_n , if $p \neq 2$, then ${}^p b_x = b_x$. If $p = 2$, then ${}^p b_x = b_x$ unless $x = w_0s_1$ or $x = w_0s_1s_2s_1$. If $x = w_0s_1$, then*

$${}^p b_{w_0s_1} = b_{w_0s_1} + b_{w_0s_1s_2s_1}.$$

Proof. Let $x \in \overline{J_1}$ and $y \in W$ such that ${}^p m_{y, x} \neq 0$.

Assume that W is of type A_n or D_n . By Lemma 2.9, if $y \in \overline{J_1}$ then y and x must belong to the same right and left 0-cells. Thus, in types A_n and D_n , Lemma 4.3 and the fact that $\overline{J_1}$ is strongly regular imply that ${}^p b_x = b_x$ for each $x \in J$.

Assume now that W is of type B_n or C_n .

Case 1. Assume that $s_1 \in L(x)$. By Lemma 2.9, we must have $s_1 \in L(y)$. By Lemma 4.3, we have $y \in \overline{J_1}$, whence y and x must belong to the same right and left 0-cells. Thus, if $x = w_0[a, 1, b]$ for some $2 \leq a, b \leq n$, we have ${}^p b_x = b_x$. Assume now that there exist $a, b \geq 2$ such that $x = w_0[a, b]$ and $y = w_0[a, 1, b]$. We show inductively on $|a - b|$ that ${}^p m_{y,x} = 0$ in type C_n or if $p \neq 2$ and ${}^p m_{y,x} = 1$ in type B_n for $p = 2$.

Base case. Suppose first that $a = b$. We proceed by induction on a and on n . If $a = b = 2$ and $n = 2$, then $x = s_1 s_2 s_1$ and $y = s_1$ and this has been computed in [JW, Section 5.4]. If $n \geq 3$, then $x \in [n, 1, n]W_{\{s_1, \dots, s_{n-1}\}}$ and $y \in [n, 1, n]W_{\{s_1, \dots, s_{n-1}\}}$. Thanks to Lemma 2.25 and the induction hypothesis on n , we conclude that ${}^p b_x = b_x$ in type C_n and ${}^p b_x = b_x + b_y$ in type B_n .

Suppose now that $a = b > 2$ and consider $({}^{a-1, a}x) {}^{a, a-1} = w_0 s_{a-1}$. By the induction hypothesis and Lemma 2.18, ${}^p m_{y,x} = 0$ in type C_n or if $p \neq 2$ and ${}^p m_{y,x} = 1$ in type B_n for $p = 2$.

Induction step. Assume now that $|a - b| \geq 1$. If $a > b$, then we have ${}^{a, a-1}x = {}^{a, a-1}[a, b]w_0 = [a - 1, b]w_0$ and ${}^{a, a-1}y = [a - 1, 1, b]w_0$ by Lemma 2.19. The induction hypothesis and Lemma 2.18 imply the result. If $a < b$, then ${}^{a, a+1}x = {}^{a, a+1}[a, b]w_0 = [a + 1, b]w_0$ and ${}^{a, a+1}y = [a + 1, 1, b]w_0$. We conclude thanks to the induction hypothesis and Lemma 2.18.

Case 2. We assume that $s_1 \notin L(x)$ and $s_1 \in R(x)$. By Lemma 2.9, we must have $s_1 \in R(y)$. Lemma 4.3 shows $y \in \overline{J_1}$. Hence y and x must belong to the same right and left 0-cells and $y = x$.

Case 3. Assume that $x = w_0 s_1$ and $y \in \overline{J_1}$. We show inductively on the rank of W that if W is of type B_n or if $p \neq 2$, then ${}^p m_{y,x} = 0$ and if W is of type C_n and $p = 2$, then ${}^p m_{y,x} = 1$. If $n = 2$, then $x = s_2 s_1 s_2$ and $y = s_2$ and this has been computed in [JW, Section 5.4]. If $n \geq 3$, then $x, y \in [n, 1, n]W_{\{s_1, \dots, s_{n-1}\}}$. By Lemma 2.25 and the induction hypothesis, we conclude ${}^p m_{y,x} = 0$ in type B_n or if $p \neq 2$ and ${}^p m_{y,x} = 1$ in type C_n and $p = 2$.

Case 4. Assume that $x = w_0 s_1$ and $y = w_0 v$ where v is of the form $s_1 s_2 s_1 [3, 1] \dots [k, 1]$ for $2 \leq k \leq n$. If ${}^p m_{y,x} \neq 0$, then b_y appears as a summand in ${}^p b_{x s_2} b_{s_2}$ by Remark 2.14. Now $x s_2 \in \overline{J_1}$ with $R(x s_2) = S \setminus \{s_2\}$, thus ${}^p b_{x s_2} = b_{x s_2}$ by **Case 2**. In particular, it means that if ${}^p m_{y,x} \neq 0$, then $y \geq_R x s_2$, whence $y \in \overline{J_1}$. Thus $y = w_0 s_1 s_2 s_1$ as in **Case 3**.

Case 5. Assume that $x = w_0 s_1 s_2 s_1$ and W is of type B_n . Let $x' = w_0 s_1 s_2$. Observe that x and x' lie in the same right cell. Thus, there is $w \in W$ such that $\mu_{x',w}^x \neq 0$. Hence, there is $v \in W$ such that ${}^p \mu_{x',w}^v \neq 0$ and ${}^p m_{x,v} \neq 0$. We then consider the decomposition of ${}^p b_{x'} {}^p b_w$ into b_z for $z \in W$. Since ${}^p b_{x'} = b_{x'}$ by **Case 2**, we must have $z \geq_R x'$, whence, either $z = w_0$ or $z \sim_R x'$. In particular, $v \sim_R x' \sim_R x$ ($v \neq w_0$ since ${}^p b_{w_0} = b_{w_0}$). Since $R(v) \subseteq R(x)$ as ${}^p m_{x,v} \neq 0$, we must have $v \sim_L x$ as well, so $v \in \{w_0 s_1, x\}$. By **Case 3**, we cannot have $v = w_0 s_1$ and ${}^p m_{x,v} \neq 0$, thus we conclude that $v = x$. Moreover, if ${}^p m_{y,x} \neq 0$, then $y \sim_R x$. Besides, $R(y) = R(x)$ and $y \leq x$ in the Bruhat order, thus $y = x$ and ${}^p b_x = b_x$. $\square$

Remark 4.5. We have not described ${}^p b_{w_0 s_1 s_2 s_1}$ in type C_n for $p = 2$ and $n \geq 3$. If $n = 3$, we have

$${}^p b_{w_0 s_1 s_2 s_1} = b_{w_0 s_1 s_2 s_1} + (v + v^{-1})b_{w_0 u},$$

where $u = s_1 s_2 s_1 [3, 1]$ and $w_0 u = s_2 s_3 s_2$ ([JW, Section 5.4]). Thus, for any $n > 3$, for $y = w_0 u \in [n, 1, n] \dots [4, 1, 4]W_{\{s_1, s_2, s_3\}}$, we have ${}^p m_{y, x} = v + v^{-1}$ by Lemma 2.25.

4.2. The p -cells coming from the submaximal cell. We are now ready to describe how the elements ${}^p b_x$ for $x \in \overline{J_1}$ distribute over two-sided p -cells. We first choose a specific element x such that ${}^p b_x = b_x$ and show that its two-sided p -cell is contained in $\overline{J_1}$ and is submaximal (Lemma 4.9).

Lemma 4.6. *Let $x \in \overline{J_1}$ such that $l(x) = l(w_0) - 1$. Then ${}^p m_{x, w} = 0$ for all $w \neq x$.*

Proof. If ${}^p m_{x, w} \neq 0$, we must have $x < w$ in the Bruhat order or $x = w$. In the first case, $l(w) > l(w_0) - 1$ and so $w = w_0$ which contradicts Remark 2.12. Thus $w = x$. $\square$

Lemma 4.7. *Let $x \in W$ such that ${}^p b_x = b_x$. Let $y \in W$. For $X \in \{L, R, J\}$, if $y \geq_X^p x$, then for all $z \in W$ such that ${}^p m_{z, y} \neq 0$, $z \geq_X x$. In particular, $y \geq_X x$.*

Proof. We consider the case $X = L$, the case $X = R$ being similar. Since $y \geq_L^p x$, there is $w \in W$ such that ${}^p \mu_{w, x}^y \neq 0$. Let $z \in W$ such that ${}^p m_{z, y} \neq 0$. Then, ${}^p m_{z, y} {}^p \mu_{w, x}^y \neq 0$ and by Equation 2.2, there is $w', x' \in W$ such that ${}^p m_{w', w} {}^p m_{x', x} {}^p \mu_{w', x'}^z \neq 0$. Since ${}^p b_x = b_x$, we must have $x' = x$, whence $z \geq_L x$.

Suppose that $y \geq_J^p x$, then there is $y_1 \in W$ such that $y \geq_L^p y_1$ and $y_1 \geq_R^p x$. Let $z \in W$ such that ${}^p m_{z, y} \neq 0$. Since $y \geq_L^p y_1$, there is $w \in W$ such that ${}^p \mu_{w, y_1}^y \neq 0$. Thus, there exist $w', y'_1 \in W$ such that ${}^p m_{w', w} {}^p m_{y'_1, y_1} {}^p \mu_{w', y'_1}^z \neq 0$. Hence, $z \geq_L y'_1$ and, by the previous case, $y'_1 \geq_R x$. Thus, $z \geq_J x$. $\square$

The following lemma describes a particular $x \in \overline{J_1}$ such that for any $y \in \overline{J_1}$, we can find $\tilde{y} \in \overline{J_1}$ such that $\tilde{y} \sim_J^p x$ and b_y is a coefficient of ${}^p b_{\tilde{y}}$.

Lemma 4.8. *Let $x \in W$ such that ${}^p b_x = b_x$ and ${}^p m_{x, w} = 0$ for all $w \neq x$. Let $y \in W$ such that $y \sim_J x$. Then there is $\tilde{y} \in W$ such that*

- (1) ${}^p m_{y, \tilde{y}} \neq 0$,
- (2) $\tilde{y} \geq_J x$,
- (3) and $\tilde{y} \sim_J^p x$.

If $x \in \overline{J_1}$, then $\tilde{y} \sim_J x$.

Proof. Since $x \leq_J y$, b_y is a summand of ${}^p b_z {}^p b_x {}^p b_{z'}$ for some $z, z' \in W$. In particular, there is $\tilde{y} \in W$ such that ${}^p m_{y, \tilde{y}} \neq 0$ and $\tilde{y} \geq_J^p x$. By Lemma 4.7,

we have $\tilde{y} \geq_J x$. On the other hand, since $y \leq_J x$, b_x is a summand of $b_z b_y b_{z'}$ for some $z, z' \in W$, whence b_x is a summand of ${}^p b_z {}^p b_y {}^p b_{z'}$. By assumption, we must have $x \geq_J^p \tilde{y}$.

Lastly, since $y \neq w_0$, we must have $\tilde{y} \sim_J x$ if $x \in \overline{J_1}$. $\square$

Lemma 4.9. *Let $x \in \overline{J_1}$ such that ${}^p b_x = b_x$. Let ${}^p J$ be the two-sided p -cell containing x . Then ${}^p J \subseteq \overline{J_1}$. If, moreover, ${}^p m_{x,z} = 0$ for all $z \neq x$, then for any $w \in W$, if $w \not\leq_J^p {}^p J$, then $w = w_0$.*

Proof. Let $y \geq_J^p x$ and $y \neq w_0$. Lemma 4.7 implies $y \geq_J \overline{J_1}$. In particular, if $y \in {}^p J$, then $y \in \overline{J_1}$. If $y = w_0$, then $x \not\leq_J y$ and thus Lemma 4.7 shows $x \not\leq_J^p y$ and $y \notin {}^p J$. Thus, ${}^p J \subseteq \overline{J_1}$.

Finally, let $w \in W$. By definition of $\overline{J_1}$, we have $w \leq_J x$ or $w = w_0$. If $w \neq w_0$, then b_x is a summand of $b_z b_w b_{z'}$ for some $z, z' \in W$, whence b_x is a summand of ${}^p b_z {}^p b_w {}^p b_{z'}$. By assumption on x , we must have $x \geq_J^p w$. Thus, if $w \not\leq_J^p x$, then $w = w_0$. $\square$

We will now describe the two-sided p -cells containing the elements of $\overline{J_1}$. Again, the submaximal cell in type A_n (and in fact any 0-cell) was already treated in [Je1, Theorem 4.33]. We include our proof for convenience of the reader. We first choose an element $x \in \overline{J_1}$ as in Lemma 4.9 and describe its two-sided p -cell J . In type B_n (if $p = 2$), we then choose another element x' with ${}^p b_{x'} = b_{x'}$ and consider its two-sided cell J' . We show that those two two-sided cells are disjoint using a generalisation of the star operation and that $J \sqcup J' = \overline{J_1}$.

Theorem 4.10 (The submaximal p -cells). *If $p \neq 2$ or if W is of type A_n or D_n , the submaximal 0-cell $\overline{J_1}$ is a two-sided p -cell. It contains the longest element of a parabolic subgroup.*

Assume that $p = 2$.

- (1) *In type C_n , the set $\overline{J_1} \setminus \{w_0 s_1 s_2 s_1\}$ forms a single two-sided p -cell. It contains the longest element of a parabolic subgroup.*
- (2) *In type B_n , the submaximal two-sided 0-cell $\overline{J_1} = w_0 J_1$ is the union of two two-sided p -cells. Order the left and right cells by*

$$(s_1 \mid s_2, \dots, s_n),$$

Then one p -cell is strongly regular of size $(n-1) \times (n-1)$ containing the elements of the form $w_0[a, 1, b]$ for $2 \leq a, b \leq n$. It contains the longest element of a parabolic subgroup. The other two-sided p -cell has shape

$$\begin{array}{c|c} 2_{1,1} & 1_{1,n-1} \\ \hline 1_{n-1,1} & 1_{n-1,n-1} \end{array}.$$

It does not contain the longest element of a parabolic subgroup.

Proof. Types A_n and D_n . Assume that W is of type A_n or D_n . Let $x = w_0 s_1 \in \overline{J_1}$. Theorem 4.4 shows ${}^p b_x = b_x$ and Lemma 4.6 implies ${}^p m_{x,w} = 0$

for all $w \neq x$. Let pJ be the two-sided p -cell containing x . Lemma 4.9 yields ${}^pJ \subseteq \overline{J_1}$. We are left to show that $\overline{J_1} \subseteq {}^pJ$. Let $y \in \overline{J_1}$. By Lemma 4.8, there is $\tilde{y} \in \overline{J_1}$ such that ${}^pm_{y,\tilde{y}} \neq 0$ and $\tilde{y} \in {}^pJ$. By Theorem 4.4, we must have $\tilde{y} = y$, and thus $y \in {}^pJ$.

Type C_n . Assume now that W is of type C_n . Let $x = w_0s_2 \in \overline{J_1}$. Theorem 4.4 guarantees ${}^pb_x = b_x$ and Lemma 4.6 shows ${}^pm_{x,w} = 0$ for all $w \neq x$. Let pJ be the two-sided p -cell containing x . Lemma 4.9 implies ${}^pJ \subseteq \overline{J_1}$.

Let $y \in \overline{J_1}$ such that $y \notin \{w_0s_1s_2s_1\}$. By Theorem 4.4, if $\tilde{y} \in \overline{J_1}$ such that ${}^pm_{y,\tilde{y}} \neq 0$, then $\tilde{y} = y$. Thus by Lemma 4.8, we must have $y \in {}^pJ$.

If $y = w_0s_1s_2s_1$ and $p \neq 2$, the same argument applies and $y \in {}^pJ$. If $y = w_0s_1s_2s_1$ and $p = 2$, then by Remark 4.5 there is $z \in W$ with ${}^pm_{z,y} \neq 0$ and $z <_J x$. Thus, Lemma 4.7 shows $y \not\leq_J^p x$ and $y \notin {}^pJ$.

Type B_n . Finally, assume that W is of type B_n . Let $x = w_0s_1 \in \overline{J_1}$. Theorem 4.4 yields ${}^pb_x = b_x$ and Lemma 4.6 shows ${}^pm_{x,w} = 0$ for all $w \neq x$. Let pJ be the two-sided p -cell containing x . Lemma 4.9 implies ${}^pJ \subseteq \overline{J_1}$. If $p \neq 2$, then for any $y \in \overline{J_1}$, there is no $\tilde{y} \neq y \in \overline{J_1}$ such that ${}^pm_{y,\tilde{y}} \neq 0$, by Theorem 4.4. Thus, Lemma 4.8 guarantees $y \in {}^pJ$.

From now on, let $p = 2$. Let $y \in \overline{J_1}$ such that $y \neq w_0[i, 1, j]$ for some $i, j \geq 2$. By Theorem 4.4, there is no $\tilde{y} \neq y \in \overline{J_1}$ such that ${}^pm_{y,\tilde{y}} \neq 0$. Thus, Lemma 4.8 yields $y \in {}^pJ$.

Now let $x' = w_0[2, 1, 2] \in \overline{J_1}$. Let ${}^pJ'$ be the two-sided p -cell containing x' . Applying Lemma 2.20 repeatedly, we deduce $x' \sim_L^p w_0[a, 1, 2]$ for any $2 \leq a \leq n$. Moreover, $w_0[2, 1, b] \sim_R^p w_0[a, 1, b]$ for any $2 \leq a, b \leq n$. Thus, ${}^pJ'$ contains all the elements of $\overline{J_1}$ of the form $w_0[a, 1, b]$ for any $2 \leq a, b \leq n$. On the other hand, Lemma 4.9 guarantees ${}^pJ' \subseteq \overline{J_1}$.

Lastly, we check that ${}^pJ' \neq {}^pJ$. It suffices to show that $x' \notin {}^pJ$. Observe that $b_{w_0s_2}$ is a summand of $b_{x'}b_{s_2s_1}$. Thus, ${}^pb_{w_0s_2}$ is a summand of ${}^pb_{x'}{}^pb_{s_2s_1}$ and $x' \leq_R^p w_0s_2$. If $x' \in {}^pJ$, then there is $y \in W$ such that $x' \geq_L^p y \geq_R^p w_0s_2$. Thus, $L(w_0s_2) \subseteq L(y)$ and $R(y) \subseteq R(x')$. By Lemma 4.9, we must have $y \in {}^pJ$, hence $y \in \overline{J_1}$. Thus, either $y = x'$ or $y = w_0s_2$. If $y = w_0s_2$, then $x' = (x')^{-1} \geq_R^p y^{-1} = y$. Thus, if $x' \in {}^pJ$, then $x' \geq_R^p w_0s_2$. So we need to show that $x' \not\geq_R^p w_0s_2$. To do so, we will check that for all $y \sim_R w_0s_2$ not of the form $w_0[i, 1, j]$ for some $i, j \geq 2$, for all $z \sim_R w_0s_2$ of the form $w_0[i, 1, j]$ for some $i, j \geq 2$ and for all $w \in W$, we have ${}^p\mu_{y,w}^z = 0$. Equivalently, we verify that for each y as before, $z = w_0[2, 1, c]$ and $w \in W$, the coefficient of b_z and the coefficient of $b_{w_0[2,1,c]}$ are equal in ${}^pb_y{}^pb_w$. We write ${}^p\mu(y, w, z)$ for the coefficient of b_z in ${}^pb_y{}^pb_w$. We proceed by induction on $l(w)$.

Base case Let $s \in S$. Assume first that $s \in R(y)$. If $y = w_0[2, b]$ for some $2 \leq b \leq n$, then $b_yb_s = (v + v^{-1})b_y$ and $b_{w_0[2,1,b]}b_s = (v + v^{-1})b_{w_0[2,1,b]}$. If $y = w_0s_2s_1$, then $b_yb_s = (v + v^{-1})b_y$. In both cases, the hypothesis is verified. Assume that $s \notin R(y)$. We use a variant of the star operation which applies to s_1 and s_2 as defined in [Je1, Definition 4.1]. It exchanges $w_0[2, 1, a]$ and $w_0[2, a]$

for any $2 \leq a \leq n$ and fixes $w_0[2, 1]$. Moreover, thanks to [Je1, Corollary 4.10], for $w, v \in \{w_0[2, 1, a], w_0[2, a]\}$, we have

$$\mu_{w, s_a}^v = \mu_{w^*, s_a}^{v^*}.$$

Let $z = w_0[2, 1, c]$ for some $2 \leq c \leq n$. If $y = w_0[2, b]$ for some $2 \leq b \leq n$, then

$${}^p\mu(y, s_b, z) = \mu_{w_0[2, b], s_b}^{w_0[2, 1, c]} + \mu_{w_0[2, 1, b], s_b}^{w_0[2, 1, c]} = \mu_{w_0[2, 1, b], s_b}^{w_0[2, c]} + \mu_{w_0[2, b], s_b}^{w_0[2, c]} = {}^p\mu(y, s_b, w_0[2, c]).$$

Assume finally that $y = w_0 s_2 s_1$. Then,

$${}^p\mu(y, s_1, z) = \mu_{w_0[2, 1], s_1}^{w_0[2, 1, c]} = \mu_{w_0[2, 1], s_1}^{w_0[2, c]} = {}^p\mu(y, s_1, w_0[2, c]).$$

Therefore the hypothesis is verified.

Induction step Let $w \in W$ such that $l(w) \geq 2$. Let $s \in R(w)$ and $w_1 = ws$. Then ${}^p b_w = {}^p b_{w_1} b_s - \sum_{v < w_1} {}^p \mu_{w_1, s}^v {}^p b_v$. Let $y \sim_R w_0 s_2$ not of the form $w_0[i, 1, j]$ for some $i, j \geq 2$. Let $z = w_0[2, 1, c]$ for some $2 \leq c \leq n$. We then have

$${}^p\mu(y, w, z) = \sum_{z' \in W} {}^p \mu_{y, w_1}^{z'} {}^p \mu(z', s, z) - \sum_{v < w_1} {}^p \mu_{w_1, s}^v {}^p \mu(y, v, z)$$

By the induction hypothesis, if ${}^p \mu_{y, w_1}^{z'} \neq 0$, then $z' \sim_R w_0 s_2$ and z' is not of the form $w_0[i, 1, j]$ for some $i, j \geq 2$. Thus applying the induction hypothesis to each v and by the initial step, we obtain

$$\begin{aligned} {}^p\mu(y, w, z) &= \sum_{z' \in W} {}^p \mu_{y, w_1}^{z'} {}^p \mu(z', s, w_0[2, c]) - \sum_{v < w_1} {}^p \mu_{w_1, s}^v {}^p \mu(y, v, w_0[2, c]) \\ &= {}^p\mu(y, w, w_0[2, c]). \end{aligned}$$

This concludes the proof that ${}^p J' \neq {}^p J$ and thus the description of the decomposition of $\overline{J_1}$. $\square$

5. SUBREGULAR AND SUBMAXIMAL CELLS IN THE EXCEPTIONAL TYPES

This section is purely computational. The left cells in types G_2, F_4 and E_6 were computed by Gibson and given in [EJ].

Remark 5.1. We only describe the subregular cells in exceptional type. For G_2 and F_4 , we also include the submaximal cells; the whole cell structure in these two types can be created with the code in [MRT], which also produced the cells displayed here. The code in [MRT] also computes the p KL basis elements, but we do not list them here, since there are too many.

Theorem 5.2 (Exceptional subregular and submaximal p -cells). *For all primes not listed below, the subregular cell remains a single two-sided p -cell. In types G_2 and F_4 , the same convention applies to the submaximal cell when it is displayed below. The submaximal cells in types E_6, E_7, E_8 are not included here.*

- (a) In type G_2 , the subregular and submaximal cells coincide. The only exceptional primes are $p = 2, 3$.

(i) For $p = 2$, this cell is the union of two two-sided p -cells, of shapes

$$\frac{2_{1,1}}{1_{1,1}} \mid \frac{1_{1,1}}{2_{1,1}} \quad \text{and} \quad 1_{2,2}.$$

(ii) For $p = 3$, this cell is the union of two two-sided p -cells, of shapes

$$\frac{2_{1,1}}{2_{1,1}} \mid \frac{2_{1,1}}{3_{1,1}} \quad \text{and} \quad 1_{1,1}.$$

For $p = 3$ both cells contain a longest element of a parabolic subgroup, while for $p = 2$ only the cell of shape $1_{2,2}$ does.

- (b) In type F_4 , the only exceptional prime for the ordinary subregular and submaximal cells is $p = 2$. Here, both the ordinary subregular and submaximal cells are the union of two two-sided p -cells, of shapes

$$\frac{1_{2,2}}{1_{2,2}} \mid \frac{1_{2,2}}{2_{2,2}} \quad \text{and} \quad 1_{2,2}.$$

In the subregular case, both contain a longest element of a parabolic subgroup, while in the submaximal case neither cell contains such an element.

- (c) In types E_6, E_7, E_8 , the ordinary subregular cell remains a single two-sided p -cell in every characteristic. Its shape is

$$E_6 : 1_{6,6}, \quad E_7 : 1_{7,7}, \quad E_8 : 1_{8,8}.$$

All of these contain longest elements of a parabolic subgroup.

Proof. Use the code in [MRT]. □

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